

Binary Carry Structure in Iterated Lacunary Series

A complete modulo-four theorem for OEIS A168362,
an exact exception criterion, and a logarithmic counting law

Research note prepared for Vladimir Reshetnikov with OpenAI

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Abstract

Let

$$F(x) = \sum_{j \geq 0} x^{2^j} = x + x^2 + x^4 + x^8 + \cdots, \quad G_m(x) = F^{\circ m}(x),$$

and let OEIS A168362 be

$$a(n) = [x^n] G_n(x).$$

The OEIS entry records two 2026 conjectural observations: the ratio $a(n+1)/a(n) \sim en$, and a sparse pattern for $a(n) \bmod 4$. This paper settles the second observation completely and strengthens it from the diagonal to the entire iteration array.

The main structural result is that

$$[x^N] G_m(x) \equiv 0 \pmod{4} \quad \text{whenever } \text{wt}_2(N) \geq 3,$$

for every $m, N \geq 1$, where $\text{wt}_2(N)$ is the number of 1's in the binary expansion of N . The coefficients at powers of two and at sums of two powers of two are then described by exact finite bit formulas. Specializing to the diagonal gives a complete classification. Besides the four initial exceptions 2, 3, 4, 6, write $n = 2^r + 2^s$ with $r > s \geq 1$ and $d = r - s$. For $r \geq 3$, one has $a(n) \equiv 2 \pmod{4}$ precisely when $d \bmod 2^s$ belongs to an explicitly computable residue set R_s . The first exceptions are

$$2, 3, 4, 6, 20, 24, 40, 68, 136, 260, 520, 1028, 1032, 2056, 2064, \dots$$

Thus the theorem confirms every exception printed in the OEIS entry and predicts all subsequent ones.

Finally, if $E(X)$ counts the exceptional indices up to X , then

$$E(X) = \kappa \log_2 X + O((\log \log X)^3), \quad \kappa = \sum_{s \geq 1} \frac{|R_s|}{2^s} = 2.323930059308106 \dots$$

All finite calculations in the article are reproduced by a standard-library Python verifier included with the source archive. The argument itself is symbolic and does not depend on the computation.

Status and proof boundary. This is an AI-assisted, unrefereed research manuscript. The congruence statements are proved in the text. The accompanying program independently checks direct truncated composition modulo 4 through degree and iterate 256, checks the published initial terms, and regenerates all residue tables. No claim of formal verification in Lean or of publication priority is made. Independent mathematical review is recommended before an OEIS submission.

[Write note, batch 73O1: filed in ProveIt's research-report collection; provenance and status in Section 1.3.]

[Write note, batch 114, 6 October 2026.] Since batch 114 the report has two Parts. Part I (Sections 1–14 and Appendices A–B) is the report described above, unchanged except for the Part headings and dated notes. Part II (Section 15; Sections 15–27 and Appendices C–D) adds the manuscript *Digit-Sum Divisibility in Lacunary Iteration* (5 October 2026): for every prime p and every integer iterate m , inverse iterates included, the coefficients of $F_p^{\circ m}$, where $F_p = \sum_j x^{p^j}$, satisfy $v_p([x^N]F_p^{\circ m}) \geq (s_p(N) - 1)/(p - 1)$. It answers Research question 13.2, re-scopes Research question 13.3, and proves Theorem 1.2 and Propositions 5.1 and 6.1 again at $p = 2$ as special cases. Its labels carry the prefix `lim:dsd:`. The write also corrected the names of cross-references: until batch 114 every lemma, proposition, corollary, definition, conjecture, remark and research question of Part I was cited as a “Theorem” (for example “Summing Theorem 3.1” for Lemma 3.1); each is now cited by its own name. No number changed.

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Part I

The modulo-four theorem: binary carry structure

1 The OEIS problem and the new results

The sequence A168362 is defined by the main diagonal of the iteration table of $F(x) = \sum_{j \geq 0} x^{2^j}$:

$$a(n) = [x^n] F^{\circ n}(x). \quad (1)$$

Its initial values are

$$1, 2, 6, 34, 280, 3010, 39984, 634040, 11704548, \dots$$

The current OEIS entry records the conjectural ratio

$$\frac{a(n+1)}{a(n)} \sim en \quad (2)$$

and the observed congruence pattern

$$a(n) \equiv 2 \pmod{4} \quad \text{at } n = 2, 3, 4, 6, 20, 24, 40, 68, 136, 260, 520, \dots, \quad (3)$$

with apparent divisibility by 4 elsewhere; see [1].

The purpose of this article is arithmetic rather than asymptotic. The ratio problem (2) remains open here. The modulo-four observation, however, is completely resolved. The result is stronger than a list of indices: it gives a short bitwise decision procedure, describes every entry of the two-dimensional iteration table modulo 4, and yields an asymptotic law for the number of exceptions.

1.1 Statement of the diagonal classification

For nonnegative integers u, v , $u \& v$ and $u | v$ denote bitwise AND and OR. Let $\text{wt}_2(u)$ be the binary Hamming weight. For $s \geq 1$, put $M_s = 2^s - 1$ and define a subset $R_s \subseteq \mathbb{Z}/2^s\mathbb{Z}$ by

$$\mathbf{1}_{d \in R_s} \equiv \mathbf{1}_{d \equiv 0 \pmod{2^s}} + \sum_{i=0}^{s-1} \mathbf{1}_{i | ((i+d) \bmod 2^s) = M_s} \pmod{2}. \quad (4)$$

This is an exact finite definition involving only $s + 2$ Boolean tests.

Theorem 1.1 (Complete modulo-four classification for A168362). *For $n > 1$, $a(n)$ is even. Moreover, $a(n) \equiv 2 \pmod{4}$ if and only if one of the following holds:*

- (i) $n \in \{2, 3, 4, 6\}$;
- (ii) $n = 2^r + 2^s$ with $r > s \geq 1$, $r \geq 3$, and $r - s \pmod{2^s} \in R_s$.

Every other $n > 1$ satisfies $a(n) \equiv 0 \pmod{4}$.

The residue sets begin

$$\begin{aligned}
 R_1 &= \emptyset, \\
 R_2 &= \{0, 2\}, \\
 R_3 &= \{0, 1, 2, 4, 6, 7\}, \\
 R_4 &= \{0, 7, 9, 10, 11, 12, 14, 15\}, \\
 R_5 &= \{0, 21, 22, 23, 27, 28, 30, 31\}, \\
 R_6 &= \{0, 51, 54, 57, 59, 60, 62, 63\}.
 \end{aligned}$$

For example, $s = 2$ gives the particularly simple family

$$a(2^r + 4) \equiv 2 \pmod{4} \iff r \text{ is even, } r \geq 3.$$

This produces 20, 68, 260, 1028, 4100, 16388, ...

[Write note, batch 7301.] Since $R_1 = \emptyset$, case (ii) of Theorem 1.1 never holds with $s = 1$: the exceptional two-bit indices all have $s \geq 2$. The condition is stated as $r > s \geq 1$ in the source and is kept.

1.2 A theorem for the whole iteration array

The diagonal theorem follows from a more general statement. Write

$$g_m(N) = [x^N] G_m(x), \quad G_m(x) = F^{\circ m}(x). \quad (5)$$

Theorem 1.2 (Binary-weight support modulo four). *For every $m, N \geq 1$,*

$$\text{wt}_2(N) \geq 3 \implies g_m(N) \equiv 0 \pmod{4}.$$

In addition, $g_m(N)$ is odd only when N is a power of two. Exact formulas for the remaining one-bit and two-bit exponents are given in Theorem 5.2.

[Write note, batch 114, 6 October 2026.] Part II proves $2^{s_2(N)-1} \mid [x^N] F^{\circ m}(x)$ for every $N \geq 1$ and every integer m , inverse iterates included ((41), the case $p = 2$ of Theorem 16.1). Since $s_2(N) = \text{wt}_2(N)$, this contains both assertions of the theorem, by a different route (scaling to integral exponential series and Legendre's formula, with no carry calculus; write note in Section 18), and it holds modulo every 2^q . The exact formulas of Theorem 5.2 are not proved again there.

This support theorem is not a numerical coincidence. It is a consequence of the fact that, modulo 4, the only cross terms that survive a 2^k -th power are balanced products of two monomials. That two-adic fact is the core of the proof.

1.3 Place in the ProveIt collection

[Write note, batch 7301, 1 October 2026.] This report is filed in ProveIt's research-report collection, in the directory

SetTheory/Cardinals/docs/reports/congruences-and-valuations/
iterated-series/a168362-lacunary-iterates-mod4.

Its README there is the guide to the files.

Provenance. The report is built from one manuscript: manuscript 49 of batch 73 (cluster O1) of ProveIt's incoming-reports intake, the archive `ProveIt_A168362_Mod4_Theorem.zip`, which arrived in commit `3a9518c52` and was placed by commit `9df4ba51a`. The manuscript names no pinned commit; it says the repository [10] was inspected on 1 October 2026. The text is printed as delivered. The write changed only the label names, which now carry the prefix `lim:`, and added the notes marked *[Write note]*. Nothing was merged into it; no other manuscript of the batch treats this sequence. The delivered bibliography also lists two general references, on formal power series [8] and on analytic combinatorics [9], which the delivered text never cites; they are kept as background.

Status. The manuscript was prepared with OpenAI. It is unrefereed, and nothing in it is formalized in Lean or any other proof assistant. The intake reran the shipped verifier on a copy (exit code 0; the three data files it writes equal the shipped ones, up to line endings). Independently of the delivered code, it composed F modulo 4 to degree and iterate 1100: the diagonal exceptions are 2, 3, 4, 6, 20, 24, 40, 68, 136, 260, 520, 1028, 1032, as Theorem 1.1 predicts, and Theorem 1.2 holds for $m \leq 300$, $N \leq 1100$. It also recomputed $a(1), \dots, a(10)$, the sets $R_1, \dots, R_8$, the tail bound of the series for κ , and the row $n = 20$ of Table 2. It did not re-derive every proof. The OEIS statements quoted here (the comments of 22 May 2026 on A168362, and the entries A112317, A122888 and A086753) were confirmed on the live entries on 1 October 2026.

Neighbouring reports. In the same collection:

- the two other reports in `iterated-series` prove congruences for compositional iterates of other series:
`a396807-self-iterating-series` and
`compositional-tree-series-congruences`;
- `a000139-binary-carries-parity` uses the same binary-carry (Kummer–Lucas) technique for a different sequence.

There is no overlap, and no statement here depends on them.

2 Formal-series setup and binary lifts

All power series below have zero constant term, so formal composition is well-defined. We work coefficientwise; therefore every assertion remains valid after truncation to any finite degree.

For a binary series $P = \sum p_n x^n \in \mathbb{F}_2[[x]]$, let

$$[P] = \sum \tilde{p}_n x^n \in \mathbb{Z}/4\mathbb{Z}[[x]]$$

denote the canonical lift, where $\tilde{p}_n \in \{0, 1\}$. Every series $H \in \mathbb{Z}/4\mathbb{Z}[[x]]$ has a unique binary decomposition

$$H = [P] + 2[Q], \quad P, Q \in \mathbb{F}_2[[x]]. \quad (6)$$

The first component records coefficient parity, while the second records the next binary digit.

Let T be Frobenius on $\mathbb{F}_2[[x]]$:

$$T(P) = P^2.$$

Because the characteristic is two,

$$F(P) = P + P^2 + P^4 + \dots = (1 + T)^{-1}P.$$

Here the inverse is the coefficientwise convergent geometric series $1 + T + T^2 + \dots$.

3 The modulo-four polarization identity

3.1 A multinomial valuation lemma

Lemma 3.1 (Only balanced pairs survive). *Let $z_1, z_2, \dots$ be commuting indeterminates of positive degree. For $k \geq 1$,*

$$\left(\sum_j z_j \right)^{2^k} \equiv \sum_j z_j^{2^k} + 2 \sum_{i < j} (z_i z_j)^{2^{k-1}} \pmod{4}. \quad (7)$$

Proof. Consider a multinomial term with exponents $e_1, e_2, \dots$ summing to 2^k . Legendre's formula gives

$$v_2 \left(\frac{(2^k)!}{\prod_j e_j!} \right) = \sum_j s_2(e_j) - 1,$$

where s_2 is binary digit sum. The valuation is zero exactly when one $e_j = 2^k$ and all other exponents vanish. The valuation is one exactly when the total binary digit sum of the nonzero exponents is two. Since their sum is a single power of two, this forces exactly two nonzero exponents, both equal to 2^{k-1} . Every other multinomial coefficient is divisible by 4. The balanced binomial coefficient has valuation one, hence is congruent to 2 modulo 4. This proves (7). $\square$

3.2 Polarizing the lacunary operator

Summing Lemma 3.1 over k gives a compact identity.

Corollary 3.2 (Lacunary polarization). *For positive-degree monomials z_j ,*

$$F \left(\sum_j z_j \right) \equiv \sum_j F(z_j) + 2 \sum_{i < j} F(z_i z_j) \pmod{4}. \quad (8)$$

A second elementary fact explains how the $2[Q]$ component propagates.

Lemma 3.3 (The high binary digit is carried unchanged). *For $P, Q \in \mathbb{F}_2[[x]]$,*

$$F([P] + 2[Q]) \equiv F([P]) + 2[Q] \pmod{4}. \quad (9)$$

Proof. The term with exponent one is $[P] + 2[Q]$. For every $k \geq 1$, $([P] + 2[Q])^{2^k} \equiv [P]^{2^k} \pmod{4}$, because every term containing $2[Q]$ has a factor divisible by 4. $\square$

4 The carry transform

The parity iterates are supported only on powers of two. It is therefore useful to encode a subset $S \subseteq \mathbb{Z}_{\geq 0}$ by

$$P_S(x) = \sum_{j \in S} x^{2^j} \in \mathbb{F}_2[[x]].$$

For $r \geq 0$, define the prefix count

$$N_S(r) = |S \cap \{0, 1, \dots, r\}|.$$

Definition 4.1 (Carry transform). The binary series $\mathcal{C}(P_S)$ is defined by

$$[x^{2^r}] \mathcal{C}(P_S) = \binom{N_S(r)}{2} \pmod{2}, \quad (10)$$

$$[x^{2^r+2^s}] \mathcal{C}(P_S) = \sum_{i=0}^s \mathbf{1}_{i \in S} \mathbf{1}_{i+r-s \in S} \pmod{2}, \quad r > s, \quad (11)$$

and all coefficients at exponents of binary weight at least three are zero.

Proposition 4.2 (Exact lift of one lacunary step). *For every S ,*

$$F([P_S]) \equiv [F(P_S)] + 2[\mathcal{C}(P_S)] \pmod{4}. \quad (12)$$

Proof. By Corollary 3.2,

$$F([P_S]) \equiv \sum_{j \in S} F(x^{2^j}) + 2 \sum_{\substack{i < j \\ i, j \in S}} F(x^{2^i+2^j}) \pmod{4}.$$

At exponent 2^r , the first sum has coefficient $N_S(r)$. Its low binary digit is $N_S(r) \bmod 2$, and its next digit is

$$\left\lfloor \frac{N_S(r)}{2} \right\rfloor \equiv \binom{N_S(r)}{2} \pmod{2}.$$

This gives (10). In the second sum, the exponent $2^r + 2^s$ occurs from a pair (i, j) precisely when $(i, j) = (s - k, r - k)$ for some $0 \leq k \leq s$. Reindexing by $i = s - k$ gives (11). No exponent with three binary 1's can occur. $\square$

5 Closed form for the parity iterates

Write

$$G_m(x) \equiv [P_m(x)] + 2[Q_m(x)] \pmod{4}, \quad P_m, Q_m \in \mathbb{F}_2[[x]]. \quad (13)$$

Starting with $P_0 = x$ and $Q_0 = 0$, Lemma 3.3 and Proposition 4.2 give

$$P_{m+1} = F(P_m), \quad Q_{m+1} = Q_m + \mathcal{C}(P_m) \text{ in } \mathbb{F}_2[[x]]. \quad (14)$$

Proposition 5.1 (Negative-binomial parity formula). *For $m \geq 1$,*

$$P_m(x) = \sum_{j \geq 0} \binom{m+j-1}{j} x^{2^j} \text{ in } \mathbb{F}_2[[x]]. \quad (15)$$

Equivalently, if

$$S_m = \{j \geq 0 : j \& (m-1) = 0\},$$

then $P_m = P_{S_m}$.

Proof. Over $\mathbb{F}_2[[x]]$, $F = (1 + T)^{-1}$. Therefore

$$P_m = (1 + T)^{-m} x = \sum_{j \geq 0} \binom{m+j-1}{j} T^j x,$$

which is (15). Lucas's theorem, or equivalently the no-carry form of Kummer's theorem, says that $\binom{m+j-1}{j}$ is odd exactly when the addition of j and $m-1$ has no binary carry, namely when $j \& (m-1) = 0$; see [7]. $\square$

[Write note, batch 114, 6 October 2026.] Part II proves $A_p(m, p^r) \equiv (-1)^r \binom{-m}{r} \pmod{p}$ for every prime p and every $m \in \mathbb{Z}$ (51), by the same operator argument with $\tau: H(x) \mapsto H(x^p)$ in place of T . At $p = 2$ and $m \geq 1$ this, with the support of P_m at powers of two, is (15) (write note in Section 19).

Combining the recurrence with the closed form gives the promised theorem for the complete array.

Theorem 5.2 (Exact array formula modulo four). *Let $m \geq 1$.*

- (a) *If $\text{wt}_2(N) \geq 3$, then $g_m(N) \equiv 0 \pmod{4}$.*
- (b) *If N is not a power of two, then $g_m(N)$ is even.*
- (c) *For $N = 2^r$,*

$$g_m(2^r) \equiv \binom{m+r-1}{r} + 2 \sum_{\ell=1}^{m-1} \binom{\#\{0 \leq j \leq r : j \& (\ell-1) = 0\}}{2} \pmod{4}. \quad (16)$$

- (d) *For $N = 2^r + 2^s$ with $r > s \geq 0$, put $d = r - s$ and*

$$E(M, B) = \#\{0 \leq t \leq M : t \& B = 0\} \pmod{2}, \quad E(M, B) = 0 \text{ for } M < 0. \quad (17)$$

Then

$$g_m(2^r + 2^s) \equiv 2 \sum_{i=0}^s E(m-2, i \mid (i+d)) \pmod{4}. \quad (18)$$

Proof. By (14), $Q_m = \sum_{\ell=0}^{m-1} \mathcal{C}(P_\ell)$; the $\ell = 0$ carry is zero. Every P_ℓ is supported at one-bit exponents, and every $\mathcal{C}(P_\ell)$ is supported at exponents of weight at most two. This proves (a) and (b). Formula (16) follows from (10) and Proposition 5.1.

For a two-bit exponent, substitute S_ℓ into (11). With $t = \ell - 1$, membership of both i and $i + d$ in S_ℓ is equivalent to

$$t \& i = 0, \quad t \& (i + d) = 0,$$

and hence to $t \& (i \mid (i + d)) = 0$. Summing $0 \leq t \leq m - 2$ gives (18). $\square$

Remark 5.3 (A logarithmic-time evaluation of E). The parity in (17) can be evaluated without enumeration. Every admissible $t \leq M$ is either the tight word $t = M$ or has a highest bit j at which $M_j = 1$ and $t_j = 0$. Hence

$$E(M, B) = \mathbf{1}_{M \& B = 0} \oplus \bigoplus_{\substack{j: M_j = 1; \\ (M \gg (j+1)) \& (B \gg (j+1)) = 0; \\ B \bmod 2^j = 2^j - 1}} 1, \quad (19)$$

where $\oplus$ denotes XOR. Thus every entry at a two-bit exponent is computable in $O(s \log m)$ bit operations.

6 Specialization to the A168362 diagonal

We now set $m = N = n$.

6.1 All noninitial diagonal terms are even

Proposition 6.1 (Diagonal parity). *For every $n > 1$, $a(n)$ is even.*

Proof. If n is not a power of two, this follows from Theorem 5.2(b). If $n = 2^r$ with $r \geq 1$, the parity is

$$\binom{2^r + r - 1}{r} \pmod{2}.$$

By Lucas's theorem it would be odd only if $r \& (2^r - 1) = 0$. But $0 < r < 2^r$ and the lowest r binary positions of $2^r - 1$ are all 1, so $r \& (2^r - 1) = r \neq 0$. $\square$

[Write note, batch 114, 6 October 2026.] Part II's Corollary 19.2 proves $p \mid A_p(N, N)$ for every prime p and every $N > 1$. At $p = 2$ it is this proposition, with a different argument at $n = 2^r$ (write note in Section 19).

6.2 Powers of two

Proposition 6.2 (The one-bit diagonal). *For $r \geq 1$,*

$$a(2^r) \equiv 2 \pmod{4} \iff r \in \{1, 2\}.$$

Thus only $n = 2$ and $n = 4$ are exceptional powers of two.

Proof. For $t = 0, 1, \dots, 2^r - 2$, put

$$N_t = \#\{0 \leq j \leq r : j \& t = 0\}.$$

By (16) and Proposition 6.1, the second binary digit of $a(2^r)$ is

$$\sum_{t=0}^{2^r-2} \binom{N_t}{2} \pmod{2}.$$

Interchange the sum over t with the choice of a pair $0 \leq i < j \leq r$:

$$\sum_{0 \leq i < j \leq r} \#\{0 \leq t \leq 2^r - 2 : t \& (i \mid j) = 0\} \pmod{2}. \quad (20)$$

The excluded value $t = 2^r - 1$ is never counted because $i \mid j > 0$. Over the complete set of r -bit values, the inner count is $2^{r-\text{wt}_2(i \mid j)}$, which is odd exactly when $i \mid j = 2^r - 1$. For $r = 1$, the pair $(0, 1)$ works; for $r = 2$, the pair $(1, 2)$ works. For $r \geq 3$, both i and j are at most $r < 2^{r-1}$, so their OR is smaller than $2^r - 1$. This proves the claim. $\square$

6.3 Two-bit exponents

For $n = 2^r + 2^s$, $r > s \geq 0$, Theorem 5.2 gives the exact criterion

$$\frac{a(n)}{2} \equiv \sum_{i=0}^s E(n-2, i \mid (i+r-s)) \pmod{2}. \quad (21)$$

The following elementary evaluation is what turns this into Theorem 1.1.

Lemma 6.3 (A nearly complete binary interval). *For $s \geq 1$ and $B \geq 0$,*

$$E(2^s - 2, B) = 1 \iff B \bmod 2^s \in \{0, 2^s - 1\}. \quad (22)$$

Proof. The complete range $0 \leq t \leq 2^s - 1$ contains $2^{s-\text{wt}_2(B \bmod 2^s)}$ values disjoint from B . This count is odd exactly when the low s bits of B are all 1. Passing from the complete range to $0 \leq t \leq 2^s - 2$ removes the all-ones value, and that removed value is disjoint from B exactly when the low s bits of B are all 0. $\square$

Proof of Theorem 1.1. Parity was proved in Proposition 6.1. Exponents with at least three binary 1's are divisible by 4 by Theorem 1.2; powers of two were settled in Proposition 6.2.

Suppose $n = 2^r + 2^s$, $r > s \geq 0$. If $s = 0$, then $n - 2 = 2^r - 1$ and (21) has the single mask $B = r$. The count over all r -bit values is odd only when $r = 2^r - 1$, which occurs only at $r = 1$. This gives $n = 3$. The remaining small case $(r, s) = (2, 1)$ gives $n = 6$ by direct substitution in (21).

Now assume $r \geq 3$ and $s \geq 1$, and put $d = r - s$. Split $0 \leq t \leq n - 2 = 2^r + 2^s - 2$ into

$$0 \leq t < 2^r, \quad t = 2^r + u, \quad 0 \leq u \leq 2^s - 2.$$

For $B_i = i \mid (i + d)$, the first block has even cardinality: both i and $i + d$ are at most r , so $B_i < 2^r - 1$ when $r \geq 3$, and the number of r -bit values disjoint from B_i is an even power of two. In the second block, bit r of B_i is zero, so the parity is $E(2^s - 2, B_i)$. By Lemma 6.3, this is one exactly when the low s bits of B_i are all zero or all one. The all-zero case occurs exactly when $i = 0$ and $d \equiv 0 \pmod{2^s}$. The all-one cases are exactly the tests in (4). Their XOR is the definition of R_s . $\square$

7 Constructing the residue sets

Definition (4) already yields an $O(s)$ membership test. For enumeration and counting, a symmetric-difference description is more convenient. For $0 \leq i \leq s$, let

$$D_{s,i} = \{(((M_s \text{ xor } i) \mid u) - i) \bmod 2^s : u \text{ is a bitwise submask of } i\}. \quad (23)$$

Here $M_s \text{ xor } i = M_s - i$ because M_s consists entirely of 1's.

Proposition 7.1 (Symmetric-difference construction).

$$R_s = \{0\} \triangle D_{s,0} \triangle D_{s,1} \triangle \cdots \triangle D_{s,s}, \quad (24)$$

where $\triangle$ denotes symmetric difference. Furthermore,

$$|R_s| \leq 1 + \sum_{i=0}^s 2^{\text{wt}_2(i)} \leq 1 + \frac{(s+1)(s+2)}{2}. \quad (25)$$

For $s \geq 3$, every nonzero $d \in R_s$ satisfies

$$d \geq 2^s - 1 - 2s. \quad (26)$$

Proof. Fix i . Put $y = (i + d) \bmod 2^s$. The condition $i \mid y = M_s$ forces all zero bits of i to be one in y , while the one bits of i are free. Thus $y = (M_s \text{ xor } i) \mid u$ with u a submask of i . This proves (24). There are $2^{\text{wt}_2(i)}$ such submasks. Since the least nonnegative integer with k one-bits is $2^k - 1$, one has $2^{\text{wt}_2(i)} \leq i + 1$, giving (25). Finally, every such y is at least $M_s - i$, so for $s \geq 3$, $y - i \geq M_s - 2s > 0$. This proves (26). $\square$

The lower bound (26) explains why only small s appear in the early exception list. Although R_s is periodic modulo 2^s , its nonzero residues lie in a short interval near the top of that modulus.

Table 1: The first residue sets and their sizes.

s	$ R_s $	R_s
1	0	$\emptyset$
2	2	0, 2
3	6	0, 1, 2, 4, 6, 7
4	8	0, 7, 9, 10, 11, 12, 14, 15
5	8	0, 21, 22, 23, 27, 28, 30, 31
6	8	0, 51, 54, 57, 59, 60, 62, 63
7	12	0, 113, 114, 116, 117, 119, 120, 121, 123, 124, 126, 127
8	12	0, 239, 241, 242, 244, 245, 248, 249, 251, 252, 254, 255

8 The infinite exception sequence

Applying Theorem 1.1 gives

$$2, 3, 4, 6, 20, 24, 40, 68, 136, 260, 520, 1028, 1032, 2056, 2064, \\ 4100, 4104, 8200, 8208, 16388, 16400, 32776, 32784, 65540, 65552, \\ 131080, 262148, 262152, 262160, 524296, 524304, 1048580, \dots$$

The terms through 520 are exactly those displayed in the OEIS comment. The next two predictions are

$$a(1028) \equiv a(1032) \equiv 2 \pmod{4},$$

followed by 2056 and 2064.

Corollary 8.1 (A compact family description). *For every $r \geq 3$,*

$$\begin{aligned} a(2^r + 4) \equiv 2 \pmod{4} &\iff r \equiv 0 \pmod{2}, \\ a(2^r + 8) \equiv 2 \pmod{4} &\iff r - 3 \equiv 0, 1, 2, 4, 6, 7 \pmod{8}. \end{aligned}$$

The corresponding statements for $2^r + 2^s$ are periodic in $r - s$ with period 2^s .

9 How many exceptions are there?

Define

$$\mathcal{E}(X) = \#\{2 \leq n \leq X : a(n) \equiv 2 \pmod{4}\}. \quad (27)$$

The residue classification yields a convergent density constant on the logarithmic scale.

Definition 9.1. Let

$$\kappa = \sum_{s \geq 1} \frac{|R_s|}{2^s}. \quad (28)$$

The series converges absolutely by (25).

Theorem 9.2 (Logarithmic counting law). *As $X \rightarrow \infty$,*

$$\mathcal{E}(X) = \kappa \log_2 X + O((\log \log X)^3). \quad (29)$$

Numerically,

$$\kappa = 2.323930059308106 \dots \quad (30)$$

Proof. First take $X = 2^L - 1$. Apart from the four isolated exceptions 2, 3, 4, 6, every exceptional index has the form $n = 2^{s+d} + 2^s$ with $s \geq 2$, $d \geq 1$, $s + d \leq L - 1$, and $d \bmod 2^s \in R_s$. Hence

$$\mathcal{E}(2^L - 1) = 4 + \sum_{s=2}^{L-2} \#\{1 \leq d \leq L - 1 - s : d \bmod 2^s \in R_s\}. \quad (31)$$

For fixed s and $Y \geq 0$, periodic counting gives

$$\#\{1 \leq d \leq Y : d \bmod 2^s \in R_s\} = \frac{|R_s|}{2^s} Y + O(|R_s|). \quad (32)$$

By (26), a summand is zero once $2^s - 1 - 2s > Y$ and $2^s > Y$. Thus only $s = O(\log L)$ contribute. Summing the error in (32) and using $|R_s| = O(s^2)$ gives $O((\log L)^3)$. Replacing $Y = L - 1 - s$ by L changes the main sum by a bounded amount because $\sum s|R_s|/2^s$ converges. The tail of (28) beyond $s = 2\lceil \log_2 L \rceil$ contributes $o(1)$ after multiplication by L . This proves

$$\mathcal{E}(2^L - 1) = \kappa L + O((\log L)^3).$$

For arbitrary $X \in [2^L, 2^{L+1})$, only exponents with top bit L can be added. Again (26) restricts their lower-bit positions to $O(\log L)$ possibilities. This is absorbed by the same error term, and $L = \log_2 X + O(1)$ proves (29).

For the decimal, summing through $s = 64$ gives

$$\sum_{s=1}^{64} \frac{|R_s|}{2^s} = \frac{10717235762314325637}{4611686018427387904} = 2.323930059308106699 \dots$$

From (25), the remaining tail is at most

$$\sum_{s>64} \frac{1 + (s+1)(s+2)/2}{2^s} = \frac{64^2 + 7 \cdot 64 + 16}{2^{65}} = \frac{285}{2^{61}} < 1.24 \times 10^{-16}.$$

□

Remark 9.3. The exceptional set has natural density zero for the trivial reason that it is contained in the integers of binary weight at most two. Theorem 9.2 is sharper: among the $\asymp (\log X)^2$ integers of binary weight two below X , only $\asymp \log X$ are exceptional.

10 Independent exact verification

[Write note, batch 7301.] By default the program writes into the `data/` directory beside its own parent directory, so a run in the collection would overwrite the shipped records; its option `-root` selects another directory. The README gives a rerun on a copy. The file `data/verification_run.txt` is the captured standard output of the recorded run, which prints the same JSON record; it is byte-identical to `data/verification.json`.

The archive contains `code/verify_a168362.py`. It uses only the Python standard library and has two deliberately separate layers.

1. It directly composes truncated dense polynomials in $\mathbb{Z}/4\mathbb{Z}[x]$ by repeated squaring, without using the carry transform. Through degree and iterate 256, it checks:

- the diagonal classification;

- divisibility by 4 at every exponent of binary weight at least three;
 - the parity formula at powers of two;
 - the exact two-bit formula (18).
2. It independently regenerates the sets R_s from the symmetric difference (24), the first exceptions through 10^7 , and the partial sum for κ .

The direct 256×256 table has SHA-256 digest

`df15e35655f2aaf2905219425b6ebcf63a2b8a8daf0144380b1e21824cf091bf.`

The program also performs direct integer composition for the 21 terms currently printed on the OEIS page and obtains an exact match.

A typical run is

```
python code/verify_a168362.py --limit 256
```

and writes the machine-readable files `data/verification.json`, `data/residue_sets.csv`, and `data/exceptional_indices_to_10m.txt`.

11 What the result says about the ratio conjecture

The congruence theorem does not prove (2); the arithmetic and asymptotic questions live at different scales. Still, a few rigorous comparisons and numerical observations help delimit the next problem.

Let $B(x) = x + x^2$ and $H(x) = x/(1 - x)$. Coefficientwise,

$$B(x) \preceq F(x) \preceq H(x).$$

Composition preserves these inequalities for series with nonnegative coefficients. Since

$$H^{\circ n}(x) = \frac{x}{1 - nx},$$

we obtain

$$a(n) \leq n^{n-1}. \tag{33}$$

The lower comparator

$$b(n) = [x^n] B^{\circ n}(x)$$

is OEIS A112317 [4]. The associated array has a word interpretation: $[x^k] B^{\circ n}(x)$ counts words of length $k - 1$ on $[n]$ in which two occurrences of i are separated by a larger letter [5]. Every word of length $n - 1$ with distinct letters is admissible, and there are $n!$ such words. Therefore

$$n! \leq b(n) \leq a(n) \leq n^{n-1}. \tag{34}$$

In particular,

$$\log a(n) = n \log n + O(n).$$

Direct exact computation through $n = 300$ gives the following diagnostics. Here

$$R_n = \frac{a(n+1)}{en a(n)}, \quad C_n = \frac{a(n)}{n^{n-2}}.$$

[Write note, batch 7301.] Table 2 is not produced by the shipped verifier, and no program that

Table 2: Numerical orientation for the unresolved ratio problem.

n	R_n	C_n
20	0.930237015501	2.532871208108
50	0.970851708171	2.548375805426
100	0.985262506048	2.558520181274
150	0.990139152504	2.566400160982
200	0.992589718536	2.572452357804
250	0.994064010570	2.577220409158
299	0.995032124887	2.581012234240

produced it was delivered; it is numerical orientation only. The intake recomputed the row $n = 20$ exactly and agrees to all printed digits; the other rows were not reproduced.

The ratio data strongly support the OEIS conjecture, but the slow drift of C_n does not distinguish a finite limiting constant from a slowly varying factor. This motivates the following deliberately cautious refinement.

Conjecture 11.1 (Refined diagonal asymptotic). *There is a positive slowly varying function $L(n)$ such that*

$$a(n) = n^{n-2}L(n), \quad \frac{L(n+1)}{L(n)} \rightarrow 1.$$

A stronger possibility is that $L(n)$ converges to a finite positive constant. Either statement implies the ratio conjecture after a standard expansion.

The exponent $n - 2$ is also suggested by parabolic iteration theory. A proof will likely require a uniform saddle-point analysis of the iteration matrix, not only a fixed-order formal conjugacy.

[Write note, batch 77P2, 2 October 2026.] A worked example of such an analysis for another diagonal of the same shape is now in the collection: the report `a139383-iterated-bell-diagonals` in its directory `generating-functions-and-asymptotics/oeis-sequence-asymptotics` (batch 77, manuscripts 53 and 28) proves, for the iterated Bell numbers $H(n, m) = n! [z^n](e^z - 1)^{om}$, an expansion of the diagonal $H(n, n+k)$ to every fixed order (OEIS A139383 and A261280 are $k = 0, 1$), by an exact residue representation on a fixed contour, an effective Fatou coordinate of the parabolic germ $e^z - 1$ and uniform cut estimates; its Part II extends this to depth proportional to n . That method is a candidate route to Conjecture 11.1, offered as a method pointer only. It does not transfer as it stands: F here is lacunary, with the unit circle as natural boundary, whereas $e^z - 1$ is entire; the quadratic and cubic Taylor coefficients, which fix the logarithmic term of the Fatou coordinate, differ (1, 0 here, $\frac{1}{2}, \frac{1}{6}$ there); and H carries the factor $n!$. No statement of this report is changed, and the conjecture remains open; the iterated-Bell report's guide also lists it as open and makes no claim about it.

12 Formalization route

The proof is well suited to a short Lean development over truncated polynomials or finitely supported series. A suggested dependency order is:

1. Define binary Hamming weight, bitwise disjointness, and the canonical lift from $\mathbb{F}_2$ coefficients to $\mathbb{Z}/4\mathbb{Z}$.

2. Prove the multinomial valuation lemma for exponent 2^k . In a first formalization, this can be replaced by an induction on repeated squaring modulo 4, avoiding a general multinomial valuation library.
3. Define the carry transform $\mathcal{C}$ on finite truncations and prove Proposition 4.2.
4. Prove $(1 + T)^{-m}$ coefficient extraction and discharge the parity condition with the existing Lucas-theorem infrastructure, or with a direct bit-carry lemma.
5. Prove Theorem 5.2 and then the three diagonal cases.
6. Define R_s as a finite set and prove equivalence of (4) and (24).
7. Formalize the counting theorem separately; it uses only elementary periodic counting and geometric-series tail bounds.

The trusted core need not include the large numerical table. The theorem is symbolic; computation is useful only as regression protection and for printing examples.

13 Further research questions and topics

Research question 13.1 (The ratio conjecture). Prove $a(n+1)/a(n) \sim en$. Determine whether $a(n)/n^{n-2}$ converges, and if so identify its limit. A useful intermediate target is a two-sided bound $n^{n-2-o(1)} \leq a(n) \leq n^{n-2+o(1)}$.

[Write note, batch 114, 6 October 2026.] Still open. Part II's congruences do not bear on it (Section 26); it is restated there as Research question 27.9, with a proposed approach.

Research question 13.2 (Higher two-adic precision). Extend the carry calculus from modulo 4 to modulo 8 and 16. At modulus 8, multinomial terms of valuation two enter, so exponents of binary weight three should appear. Is there a general theorem that modulo 2^q , every coefficient at an exponent of binary weight greater than q vanishes? The balanced-partition analysis suggests such a filtration, but carries among lower layers must be controlled.

[Write note, batch 114, 6 October 2026: answered.] Yes, for every $q \geq 1$. Part II proves $2^{s_2(N)-1} \mid [x^N]F^{\circ m}(x)$ for every $N \geq 1$ and every integer m , inverse iterates included ((41), the case $p = 2$ of Theorem 16.1). So if $\text{wt}_2(N) > q$, then $\text{wt}_2(N) - 1 \geq q$ and $g_m(N) \equiv 0 \pmod{2^q}$. The proof does not extend the carry calculus: it rescales F to an integral exponential series and uses Legendre's formula (Section 18; write note there). What the layers of weight at most q carry modulo 2^q is not determined; Part II's Research question 27.1 asks for it on the diagonal modulo 8 and 16.

Research question 13.3 (Odd-prime analogues). For a prime p , replace F by

$$F_p(x) = \sum_{j \geq 0} x^{p^j}.$$

Develop the analogue modulo p^2 . The multinomial valuation argument should restrict surviving exponents to base- p digit sum at most two, but the carry operator has $p - 1$ nontrivial digits rather than one.

[Write note, batch 114, 6 October 2026: re-scoped.] The suggestion in the second sentence is false for every odd prime p . Part II proves $[x^{2p-1}]F_p(F_p(x)) = p$ (Proposition 19.1). Indeed, $F_p(F_p(x)) = \sum_{k \geq 0} F_p(x)^{p^k}$, and since $p^2 > 2p - 1$ only the terms $k = 0, 1$ reach degree $2p - 1$; the first, $F_p(x)$, has no monomial x^{2p-1} because $p < 2p - 1 < p^2$; in the second, $F_p(x)^p$, the only way to reach degree $2p - 1$ is one factor x^p and $p - 1$ factors x , and the factor x^p can be chosen in p ways. So the coefficient is $p \not\equiv 0 \pmod{p^2}$, at the exponent $2p - 1 = 1 \cdot p + (p - 1)$, whose base- p digit sum is $p > 2$. (At $p = 2$ the digit sum is 2, and there is no conflict with Part I.) The correct support statement is Theorem 16.1 with $q = 2$: a coefficient of any integer iterate of F_p can be nonzero modulo p^2 only if $N \equiv 1 \pmod{p - 1}$ and $s_p(N) \leq p$, that is, only if $s_p(N) \in \{1, p\}$. The layer $s_p(N) = p$ is occupied: $A_p(2, p^r + p - 1) \equiv p \pmod{p^2}$ for every $r \geq 1$ (Theorem 20.4). What remains open is the rest of the question, an exact modulo- p^2 calculus analogous to the carry transform of Definition 4.1: Part II determines the first residue at digit sum p for the second iterate only (Theorem 20.1), and its Research question 27.3 asks for higher iterates. This corrects a suggestion in a research question; no theorem of Part I is affected.

Research question 13.4 (Automatic and regular structure). Study the sequence $|R_s|$ and the two-parameter predicate $d \bmod 2^s \in R_s$. Is the resulting set two-automatic in a suitable paired encoding of (s, d) ? Can κ be represented as the value of a rational or algebraic generating function, rather than only as the rapidly convergent series (28)?

[Write note, batch 114, 6 October 2026.] Still open. Part II proves a different automaticity statement: for each fixed prime p , precision q and iterate $m \geq 0$, the row $N \mapsto [x^N]F_p^{om} \bmod p^q$ is p -automatic (Theorem 23.4). It says nothing about R_s , the predicate $d \bmod 2^s \in R_s$ or κ , and nothing about the diagonal.

Research question 13.5 (Other diagonals). Use Theorem 5.2 to classify

$$[x^n]F^{o(n+c)}(x) \pmod{4}$$

for fixed integer offsets c . The adjacent OEIS sequences A168365 and A168366 correspond to $c = -1$ and $c = 1$ [2, 3]. The general two-bit formula already solves the computational part; the goal is to simplify it to residue families and counting constants analogous to R_s and κ .

[Write note, batch 114, 6 October 2026.] Advanced in one direction only. Part II's bound $2^{s_2(N)-1} \mid [x^N]F^{om}(x)$ holds for every choice $m = m(N)$ (Theorem 16.1), so every shifted diagonal, A168365 and A168366 included, vanishes modulo 2^q at exponents of binary weight greater than q . The residue families and counting constants asked for here remain open. On the examples printed in the OEIS entries of A168365 and A168366 see Remark 15.1.

Research question 13.6 (Combinatorial model). Find a direct combinatorial interpretation of the coefficients of F^{om} . For $x + x^2$, the word model behind A122888 makes lower bounds and probabilistic questions natural. A model in which powers 2^j represent allowed arities may expose the saddle point and make the ratio conjecture more tractable.

Research question 13.7 (Distinct multinomial coefficients). A separate OEIS problem inspected during this project is A086753, the number of distinct trinomial coefficients in row n , with conjectural asymptotic $(n^2 + 6n - 12)/12$ [6]. Equality of two coefficients reduces to an interval-product identity between factorial ratios. Controlling the total number of nontrivial collisions, rather than only individual collision families, remains an attractive Diophantine-asymptotic problem.

[Write note, batch 7301.] The last question is an aside: A086753 has no connection with A168362 or with the methods of this report. The conjectural asymptotic $(n^2 + 6n - 12)/12$ in that OEIS entry is Vladimir Reshetnikov's.

14 Conclusion

The sparse modulo-four pattern in A168362 is governed by a simple mechanism: modulo 4, a 2^k -th power remembers only pure powers and balanced pairs. After iteration, this forces the second binary coefficient digit onto exponents of Hamming weight at most two. Lucas’s theorem then turns the parity orbit into a no-carry set, and a finite bit count yields the complete diagonal classification.

The outcome is both exact and scalable: it proves the observed OEIS pattern, predicts every future exception, gives a logarithmic counting law with an explicit constant, and applies to every entry of the iteration array. The remaining asymptotic ratio problem is untouched by the congruence proof, but the exact arithmetic structure and the reproducible computations provide a cleaner foundation for that next attack.

Part II

Digit-sum divisibility for every prime power and every integer iterate

15 Provenance, scope and notation of Part II

[*Write note, batch 114, 6 October 2026.*] Part II was added in the write of batch 114 of ProveIt’s incoming-reports intake. It is built from one manuscript, which answers Research question 13.2 of Part I for every modulus 2^q and every prime, and shows that the suggestion in Research question 13.3 fails for every odd prime. Sections 16 to 27 and Appendices C and D print it as delivered, apart from the label prefix `lim:dsd:`, the edits listed in Section 15.5, and the notes marked [*Write note, batch 114*]. This section is the write’s.

15.1 Provenance

The manuscript is the archive `Digit_Sum_Divisibility_in_Lacunary_Iteration.zip` (inner directory `lacunary_digit_divisibility/`), which arrived in commit `e4d5dcf9e` (“New research reports”, 5 October 2026; it is not part of a session bundle) and was placed in this report by commit `99053b5d1`. Its article is titled *Digit-Sum Divisibility in Lacunary Iteration: All prime powers, explicit first residues, finite automata, and p-adic iteration* (23 pages as delivered, dated October 5, 2026). Its title page reads “Prepared for Vladimir Reshetnikov” and “OpenAI ChatGPT”, and its PDF metadata names “OpenAI ChatGPT; prepared for Vladimir Reshetnikov” as author.

Its source audit, shipped as `02-digitsum-SOURCE_AUDIT.txt`, pins the repository at commit

`3412ae0741b95b3fb6d7a43477b7fac7723c36d8`

and this report’s `article.tex` at blob

`46147fca8f8fb912f7132d504b21217c0c248844`.

That blob is the text of Part I exactly as it stood until this write: the manuscript read Part I as printed here, without the notes dated 6 October 2026. Its checksum list (15 entries) verified at placement.

The files of the delivery are shipped with the prefix `02-digitsum-` (Table 3); the delivered `article.tex`, `article.pdf`, `README.txt` and `SHA256SUMS.txt` are not shipped and survive in the arrival commit. The text of Part II names the delivered files by their delivery paths, as the manuscript does.

Delivery path	Shipped path
SOURCE_AUDIT.txt	02-digitsum-SOURCE_AUDIT.txt
code/verify.py	code/02-digitsum-verify.py
code/leading_residue.py	code/02-digitsum-leading_residue.py
code/cartier.py	code/02-digitsum-cartier.py
code/make_figure.py	code/02-digitsum-make_figure.py
data/verification.json	data/02-digitsum-verification.json
data/valuation_grid.json	data/02-digitsum-valuation_grid.json
data/leading_residue_verification.json	data/02-digitsum-leading_residue_verification.json
data/cartier_verification.json	data/02-digitsum-cartier_verification.json
data/document_qa.json	data/02-digitsum-document_qa.json
figures/valuation_grid.pdf	figures/02-digitsum-valuation_grid.pdf
figures/valuation_grid.png	figures/02-digitsum-valuation_grid.png
article.tex, article.pdf	not shipped (this Part replaces them)
README.txt, SHA256SUMS.txt	not shipped (arrival commit e4d5dcf9e)

Table 3: Delivered files of Part II and their shipped names. The shipped files are byte-identical to the delivery.

15.2 The manuscript’s title page

The manuscript’s title page carries the following result box, abstract and status statement, printed here as delivered.

For every prime p , every integer iterate m , and every supported degree N ,

$$v_p \left([x^N] \left(\sum_{j \geq 0} x^{p^j} \right)^{om} \right) \geq \frac{s_p(N) - 1}{p - 1}.$$

At $p = 2$, this answers the higher-precision question in ProveIt’s A168362 report for every modulus 2^q . At odd primes, it gives the correct replacement for the proposed digit-sum cutoff.

Abstract. We prove a digit-sum divisibility theorem for all compositional iterates of $F_p(x) = \sum_{j \geq 0} x^{p^j}$. The proof conjugates the problem to integral exponential generating functions and uses the set-partition composition formula. It applies to inverse iterates and arbitrary compositions of weighted p -power seeds. A separate multinomial argument gives an explicit digit-polynomial formula for the first possible nonzero residue of the second iterate. We deduce stability under moving the highest nonzero digit and three infinite families attaining the bound. Classical Fuss–Catalan inverses certify optimality for the larger composition-and-inversion class. We also prove sparse-support bounds, a quantitative density-one divisibility law, and a constructive finite-automaton theorem for each fixed iterate modulo every prime power. Finally, coefficientwise p -adic interpolation gives a continuous iteration group and a digit-sensitive sufficient period in the iteration parameter. All proofs are included, together with exact verification programs and a source audit.

Research status. This is an AI-assisted research manuscript with complete mathematical arguments and reproducible exact checks. It has not been peer reviewed or verified in a proof assistant. The contribution is stated relative to the inspected repository report; historical publication priority is not certified. The real-growth ratio conjecture for A168362 remains open here.

15.3 What Part II proves, and what it proves again

Table 4 sets Part II against Part I. Three statements of Part I are proved again at $p = 2$ as special cases of all-prime theorems; each is printed once, in Part I, and Part II's route is recorded in a write note at the place where Part II proves the general statement. No result of Part I is used as a premise in Part II, and no result of Part II is used in Part I. Part II does not re-prove Part I's exact modulo-four formulas at two-bit exponents (Theorem 5.2(c),(d)), the diagonal classification (Theorem 1.1) or the counting law (Theorem 9.2); it credits them.

Part II	Relation to Part I
Support in $N \equiv 1 \pmod{p-1}$ and $v_p(A_p(m, N)) \geq (s_p(N) - 1)/(p - 1)$ for every prime p and every $m \in \mathbb{Z}$ (Theorems 16.1 and 18.3)	new. At $p = 2$ it answers Research question 13.2 for every q , and it contains Theorem 1.2 (second route; write note in Section 18), extended to $m \leq 0$.
$A_p(m, p^r) \equiv (-1)^r \binom{-m}{r} \pmod{p}$ (51)	at $p = 2$, $m \geq 1$, together with the support statement, this is Proposition 5.1, by the same Frobenius-operator mechanism (write note in Section 19).
$p \mid A_p(N, N)$ for $N > 1$ (Corollary 19.2)	at $p = 2$ this is Proposition 6.1 (the manuscript says so; write note in Section 19).
$[x^{2p-1}]F_p(F_p(x)) = p$ (Proposition 19.1)	shows that the suggestion in Research question 13.3 fails for every odd p ; the question is re-scoped there.
First normalized residue of the second iterate, highest-digit stability, three sharp families (Theorems 20.1, 20.3 and 20.4)	new. At $p = 2$, $m = 2$ and weights 0, 1 the residue is also given by Theorem 5.2(c),(d); both agree with direct composition. The family (62) at $p = 2$, $A_2(2, 2^r + 1) \equiv 2 \pmod{4}$, is the case $s = 0$ of (18) at $m = 2$.
Fuss-Catalan sharpness in the class $\mathcal{C}_p^1$ (Theorem 21.1)	new as an application; the inverse is classical.
Candidate counts and a density-one valuation bound (Theorems 22.1 and 22.2)	new; at $p = 2$, $q = 2$ the count is coarser than the counting law Theorem 9.2 for the diagonal, as the manuscript says.
Automaticity of each fixed row modulo p^q (Theorem 23.4)	new; a different question from Research question 13.4, which stays open.
p -adic iteration group and sufficient periods (Theorems 24.1 and 24.2)	new; no counterpart.

Table 4: What Part II proves, against Part I.

15.4 Reading conventions

Part II keeps the manuscript's symbols; no symbol is renamed and no normalization is changed. Several letters mean different things in the two Parts, and a few are overloaded inside the manuscript. Table 5 fixes the reading. Wherever Part II says “the ProveIt report”, “the repository report”, “the earlier report”, “the previous repository report”, “the source report” or “the inspected repository report”, it means Part I, and its “Question 13.2” and “Question 13.3” are Research questions 13.2 and 13.3.

15.5 Edits made by the write

- Every label of the manuscript carries the prefix `lim:dsd::`; the 84 delivered names are otherwise unchanged.

Notion	Part I	Part II
seed	$F = \sum_j x^{2^j}$	$F_p = \sum_j x^{p^j}; F_2 = F$
iterates	$G_m = F^{\circ m}, m \geq 1$	$F_p^{\circ m}$ for all $m \in \mathbb{Z}$ ($m < 0$: iterates of the compositional inverse; $F_p^{\circ 0} = x$)
coefficients	$g_m(N)$	$A_p(m, N); A_2(m, N) = g_m(N)$ for $m \geq 1$
diagonal	$a(n) = g_n(n)$	$a(N) = A_2(N, N); b_p(N) = A_p(N, N)$. Not Part I's $b(n)$ of (34), which is A112317, from iterates of $x + x^2$
binary weight	$\text{wt}_2(N)$	$s_2(N)$, the same number; s_p is the base- p digit sum
Frobenius	$T(P) = P^2$ on $\mathbb{F}_2[[x]]$	$\tau: H(x) \mapsto H(x^p)$ on additive series over $\mathbb{F}_p$. On $\mathbb{F}_2[[x]]$, $P^2 = P(x^2)$, so $T = \tau$ there, and $(1 + T)^{-1} = (1 - \tau)^{-1}$. $\mathcal{T}_p$ is a scaling, not a Frobenius (44)
$\mathcal{C}$	carry transform (Definition 4.1)	$\mathcal{C}_p, \mathcal{C}_p^1$: a class of series; $C_p(x)$: the inverse of B_p ; C : the substitution $H \mapsto H \circ F_p$; $C_{p,q}(L)$: a count. None is the carry transform
R	$R_s \subseteq \mathbb{Z}/2^s$ (residue sets); R_n (ratio, Table 2)	R without subscript is local: the cutoff digit sum $1 + (q - 1)(p - 1)$ in Theorem 22.1, the lower part $0 \leq R < p^r$ in Theorem 20.3, a finite ring in Lemma 23.3
P, Q	P_m, Q_m (binary digits of G_m), P_S	$P_N(z)$ (Mahler polynomial, Theorem 24.1); P_R (digit polynomial of R)
d	$d = r - s$	$d_p(N) = (N - 1)/(p - 1)$; digits d_i of N ; generator counts d_m ; $d = d_p(N)$ in Theorem 24.2
S	S_m (no-carry set), $N_S(r)$	$S_i(z) = z + z^p + \cdots + z^{p^i}$; $\mathcal{S}_p$ (support space); $\mathcal{S}(B)$ (Mahler sum)
s	index of R_s ; $n = 2^r + 2^s$	$s_p(N)$; a digit sum s in Sections 20 and 22
B, H	$B(x) = x + x^2$, $H(x) = x/(1 - x)$ (Section 11, (33))	$B_p(x) = x + x^p$, so $B_2 = B$; H, K, B are generic series elsewhere
L	$L(n)$ slowly varying (Conjecture 11.1)	L : number of base- p digits; Λ_r : Cartier operators; $\mathcal{L}_p$: a linear functional
D, E, w	$E(M, B)$, $\mathcal{E}(X)$; no D, w	$D = C - I$; $D_p(m, N)$ (scaled integer); $w_p(N) = (s_p(N) - 1)/(p - 1)$ and w in Theorems 20.1, 20.3 and 24.2; no E
questions	Research questions 13.1–13.7	Research questions 27.1–27.10 (the manuscript's 1–10)

Table 5: Reading conventions of Part II against Part I.

- Numbering follows the report: the manuscript’s Section n is Section $n + 15$, so its Theorem $n.k$ is Theorem $(n + 15).k$; its appendices A and B are Appendices C and D; its research questions 1–10 are Research questions 27.1–27.10; equations are numbered consecutively through the whole report, as in Part I, instead of within sections.
- The bibliography is merged: the manuscript’s entries for A168362, A168365, A168366 and for Flajolet–Sedgewick are Part I’s entries [1, 2, 3, 9]; its annotations on the two neighbouring OEIS entries are discussed in Remark 15.1, and its A168362 entry adds the dates of the entry (November 23, 2009) and of its revision (May 22, 2026) and that it was inspected on October 5, 2026; its other entries are printed with the keys `dsd...` [11, 12, 13].
- The title page is printed in Section 15.2; its coloured box is a plain frame. The manuscript’s colours, header, paragraph and list spacing are not used. The figure is included from its shipped name.
- The preamble gains the manuscript’s macros and two packages (`graphicx`, `mathrsfs`), and `cleveref` is told the names of the environments that share the theorem counter, so that a lemma is cited as a lemma in both Parts (in Part I it was cited as a “Theorem” until this write).
- The notes marked *[Write note, batch 114]* and this section are added. No mathematical statement, proof or table of the manuscript is changed.

15.6 The OEIS entries A168365 and A168366

Remark 15.1 (The examples of A168365 and A168366). The manuscript identifies A168365 and A168366 “using their definitions, listed terms, and programs” and notes that “the prose expansion examples have an extraneous factor” (its source audit; its bibliography says that an “inconsistent factor in the prose example is not used”). The intake confirms this on the live entries (revision #2 of each, 30 March 2012, read on 5 October 2026). The EXAMPLE section of each reads

`define F_{n}(x) = F_{n-1}(x*F(x)) as the n-th iteration of F(x)`

with $F(x) = x + x^2 + x^4 + \dots$. Read literally with $F_0(x) = x$, this gives $F_1(x) = xF(x) = x^2 + x^3 + x^5 + \dots$, which is not F . The rows printed beneath it are nevertheless the coefficient lists $[x^1], [x^2], \dots$ of the plain iterates F^{on} : the intake recomputed all 114 printed entries of A168365 and all 110 of A168366 by direct composition, and they agree. The DATA and the PARI programs (which substitute the series into itself, `G=subst(S,x,G)`) use plain composition: the 19 terms of A168365 and the 18 of A168366 equal $A_2(N - 1, N)$ and $A_2(N + 1, N)$, and A168362’s own example defines the iterates by plain composition. So the defect is the one line defining the recursion; the sequences are those of (35), and every listed term is divisible by $2^{s_2(N)-1}$, as Theorem 16.1 requires. This is recorded only; nothing was submitted to OEIS. OEIS data are available under CC BY-SA 4.0.

15.7 What the intake checked

The intake reran the three delivered verifiers on a copy in the delivered layout: `verify.py` passes (4236 support, valuation and integrality checks, 336 Bell comparisons, 750 Fuss–Catalan valuations, 30 OEIS fixtures, the odd-prime identity at $p = 3, 5, 7, 11$), `leading_residue.py -verify` passes 1204 coefficient checks with the stability and family checks, and `cartier.py -verify` passes 2052 comparisons; each takes about a second, and the regenerated JSON files equal the shipped ones up

to line endings. The two query examples of the delivered README reproduce (normalized residue 1 at $p = 3$, $N = 51$; $A_2(2, 2^{50} + 6) \equiv 4 \pmod{8}$). `make_figure.py` was not run.

Independently of the delivered code, the intake composed the series with its own exact integer code: 3340 coefficients ($p = 2$ with $1 \leq m \leq 6$, $N \leq 160$, and $m = -1, -2, -3$, $N \leq 60$; $p = 3$ with $m \leq 5$ and $m = -1, -2$; $p = 5$; $p = 7$) satisfy the support and valuation statements of Theorem 16.1 without exception; its own implementation of (55) agrees with $A_p(2, N)/p^w \pmod{p}$ in 160, 80, 40 and 20 cases for $p = 2, 3, 5, 7$; $[x^{2p-1}]F_p(F_p(x)) = p$ for $p = 2, 3, 5, 7$; the families (62), (63) have valuations 1 and 2 where in range; the Fuss–Catalan valuation identity holds for 60 values of k at each of $p = 2, 3, 5$; the counts 1350 and 6195, the exponential coefficients 1, 1, 3, 315 and $A_2(2, 7) = 8$ are right. The write read every proof and re-derived those of Theorems 16.1, 18.3, 20.1, 20.3, 21.1, 22.1, 22.2, 23.4, 24.1 and 24.2, Proposition 19.1, and Corollary 19.2 and of (51) and (62); it found no error. The families (63) and (64) were checked numerically only.

The citations were checked: Theorem 1.8 of [13] is the kernel criterion for automaticity, and the congruence (73) is the one used in its proof of Proposition 1.9 (numbering of arXiv:1310.8635v2, which matches the manuscript's; the journal version was not read). The arXiv record of [12] matches the printed authors, title, journal and DOI, and concerns the composition of exponential generating functions over rings; Part II proves the facts it uses (Lemma 17.3). The manuscript's statements about neighbouring reports (Section 26) were checked against the reports and are correct.

16 The question and the results

Let p be a prime and work with formal power series, so no analytic convergence hypothesis is involved. Set

$$F_p(x) = \sum_{j \geq 0} x^{p^j}, \quad A_p(m, N) = [x^N]F_p^{\circ m}(x), \quad F_p^{\circ 0}(x) = x. \quad (35)$$

For $m < 0$, the superscript denotes iteration of the formal compositional inverse. This inverse exists over $\mathbb{Z}$ because the linear coefficient of F_p is one. Ordinary powers and compositional powers will always be distinguished: $F_p(x)^k$ is an ordinary product, whereas $F_p^{\circ k}$ is an iterate.

The sequence A168362 is the binary diagonal

$$a(N) = A_2(N, N) = 1, 2, 6, 34, 280, 3010, 39984, 634040, 11704548, \dots \quad (36)$$

The nearby sequences A168365 and A168366 are $A_2(N-1, N)$ and $A_2(N+1, N)$, respectively [1, 2, 3]. The live A168362 entry records Paul D. Hanna's modulo-four observation and the conjectural ratio $a(N+1)/a(N) \sim eN$.

16.1 The precise repository question

The ProveIt report *Binary Carry Structure in Iterated Lacunary Series* proves a complete modulo-four description of the iteration array and a finer classification of the exceptional diagonal indices [11]. Its Question 13.2 asks whether, modulo 2^q , every coefficient whose exponent has more than q binary ones vanishes. Its Question 13.3 proposes an odd-prime analogue, suggesting that modulo p^2 the surviving exponents should have base- p digit sum at most two.

We answer the first question for all q . The second suggestion requires correction: the cutoff is p , rather than two. The small exact identity

$$[x^{2p-1}]F_p(F_p(x)) = p \quad (37)$$

already demonstrates the issue. Since $2p - 1 = (1(p - 1))_p$, its digit sum is p . This corrects a prediction in a research question; it does not contradict a proved theorem of the earlier report.

[Write note, batch 114, 6 October 2026.] The two questions are Part I's Research questions 13.2 and 13.3; dated notes there record the answer and the re-scoping. "The ProveIt report", "the earlier report" and "the repository report" mean Part I throughout Part II (Section 15.4).

16.2 Main theorems at a glance

Write $s_p(N)$ for the sum of the base- p digits of N , and set $v_p(0) = +\infty$. At degrees $N \equiv 1 \pmod{p-1}$, define

$$d_p(N) = \frac{N-1}{p-1}, \quad w_p(N) = \frac{s_p(N)-1}{p-1}. \quad (38)$$

Both are nonnegative integers. The restriction on N is automatic for every nonzero coefficient of every iterate in (35).

Theorem 16.1 (Uniform digit divisibility). *For every prime p , every $m \in \mathbb{Z}$, and every $N \geq 1$,*

- (i) $A_p(m, N) = 0$ unless $N \equiv 1 \pmod{p-1}$;
- (ii) at a supported degree,

$$\frac{N!}{p^{d_p(N)}} A_p(m, N) \in \mathbb{Z}, \quad v_p(A_p(m, N)) \geq w_p(N). \quad (39)$$

Consequently a coefficient can be nonzero modulo p^q only if

$$N \equiv 1 \pmod{p-1}, \quad s_p(N) \leq 1 + (q-1)(p-1). \quad (40)$$

The bound is independent of m . It therefore applies equally to a fixed row, to the diagonal, to a shifted diagonal, or to an arbitrary choice $m = m(N) \in \mathbb{Z}$. At $p = 2$ the statement is particularly simple:

$$2^{s_2(N)-1} \mid A_2(m, N). \quad (41)$$

The following table distinguishes the extensions proved here from their scope. The word "fixed" in the automaton row is essential.

Result	Scope	Reference
Digit divisibility	All integer iterates; weighted seeds and their inverses	Theorems 16.1 and 18.3
First normalized residue	Exact finite digit formula for $m = 2$	Theorem 20.1
Sharpness	Three families for $F_p^{\circ 2}$; all depths in the larger class	Theorems 20.4 and 21.1
Sparse support	Every choice of iteration index $m(N)$	Theorems 22.1 and 22.2
Finite automata	Fixed $m \geq 0$ and fixed modulus p^q	Theorem 23.4
p -adic time	Coefficientwise continuous iteration; sufficient periods	Theorems 24.1 and 24.2

16.3 What is new relative to the inspected sources

The main application answers an explicit question of the repository report. The digit-polynomial residue formula, its highest-digit stability, and the fixed-iterate automaton construction provide additional precise extensions. We prove the needed facts directly rather than use the earlier report's modulo-four arguments as premises.

The tools are classical. Composition and inversion of Hurwitz series are standard; Barbero, Cerruti and Murru discuss the relevant operations [12]. Legendre's factorial formula, Wilson's theorem, formal Lagrange inversion and finite-difference interpolation are also established tools. The prime-power congruence used in the automaton argument appears in the proof of Proposition 1.9 of Rowland and Yassawi [13]. We do not claim novelty for those facts, for the Fuss–Catalan valuation used as a sharpness certificate, or for the general idea of p -adic iteration.

The source audit in Section 26 specifies the repository snapshot and the bounded literature search. No claim is made to have settled the ratio conjecture, classified every higher-modulus diagonal residue, or proved the diagonal sequence automatic.

17 Factorial valuations and integral exponential series

The divisibility proof needs two elementary facts. We include their proofs to make the arithmetic boundary explicit.

Lemma 17.1 (Legendre's formula and carries). *For every $n \geq 0$,*

$$v_p(n!) = \sum_{j \geq 1} \left\lfloor \frac{n}{p^j} \right\rfloor = \frac{n - s_p(n)}{p - 1}. \quad (42)$$

For nonnegative integers $b_1, \dots, b_r$, $s_p(\sum_i b_i) \leq \sum_i s_p(b_i)$, with equality exactly when their base- p addition involves no carries.

Proof. Each multiple of p^j in $1, \dots, n$ contributes one further factor of p , proving the first equality. If $n = \sum_i n_i p^i$, then

$$\sum_{j \geq 1} \left\lfloor n/p^j \right\rfloor = \sum_i n_i (1 + p + \dots + p^{i-1}) = \frac{\sum_i n_i (p^i - 1)}{p - 1}.$$

This gives the second equality. Replacing p units in one digit column by one unit in the next reduces the total digit sum by $p - 1$. Successively resolving carries proves the remaining assertions. $\square$

Definition 17.2. A formal power series $H(x)$ with rational coefficients is an *integral Hurwitz series* if it can be written

$$H(x) = \sum_{n \geq 0} h_n \frac{x^n}{n!}, \quad h_n \in \mathbb{Z}.$$

Equivalently, all its formal derivatives at zero are integers.

The word “integral” here concerns the exponential coefficients h_n . The ordinary coefficients $h_n/n!$ need not be integers. For example, $x + x^2/2 + x^4/8$ is an integral Hurwitz series.

Lemma 17.3 (Set-partition composition). *Let $H(x) = \sum h_n x^n/n!$ and $K(x) = \sum k_n x^n/n!$ be integral Hurwitz series, with $K(0) = 0$. Then $H \circ K$ is an integral Hurwitz series. For $n \geq 1$,*

$$n! [x^n] H(K(x)) = \sum_{\pi \in \Pi_n} h_{|\pi|} \prod_{B \in \pi} k_{|B|}, \quad (43)$$

where Π_n is the set of set partitions of $\{1, \dots, n\}$. If $H(0) = 0$ and $h_1 = 1$, its compositional inverse is also integral Hurwitz.

Proof. In the term $h_r K(x)^r / r!$, select the sizes of r nonempty blocks. After multiplication by $n!$, the multinomial factor assigns the n labels to the ordered blocks; division by $r!$ forgets their order. This gives (43). Every summand is an integer.

To construct the inverse K , impose $H(K(x)) = x$. The linear equation gives $k_1 = 1$. For $n > 1$, the partition with one block contributes k_n , the partition into singletons contributes h_n , and every other partition uses only k_j with $j < n$. Thus k_n is the negative of an integer expression in previously constructed coefficients. Induction proves the claim. Uniqueness is the usual triangular uniqueness of formal compositional inversion. $\square$

The same proof works over $\mathbb{Z}_p$ with “integral” meaning p -integral. Over a general coefficient ring one can define the operation through the set-partition polynomials themselves. We need only the concrete integer and p -adic versions, so no division by a nonunit is performed inside a ring of positive characteristic.

18 A composition class controlled by digit sums

Let $\mathcal{S}_p$ be the formal power series with rational coefficients and zero constant term supported in degrees $1 \pmod{p-1}$. When $p = 2$, this means every positive degree. Define a linear scaling on this space by

$$\mathcal{T}_p f(x) = \sum_{\substack{N \geq 1 \\ N \equiv 1 \pmod{p-1}}} [x^N] f(x) p^{-d_p(N)} x^N. \quad (44)$$

Only integer powers of p occur in this definition; no ramified field extension is needed.

Lemma 18.1 (Scaling commutes with composition). *If $f, g \in \mathcal{S}_p$, then $f \circ g \in \mathcal{S}_p$ and*

$$\mathcal{T}_p(f \circ g) = (\mathcal{T}_p f) \circ (\mathcal{T}_p g). \quad (45)$$

The same support restriction is preserved by compositional inversion when the linear coefficient is nonzero.

Proof. A contribution with outer degree k and inner degrees $j_1, \dots, j_k$ has $N = \sum_i j_i$. Since all these degrees are one modulo $p-1$, $N \equiv k \equiv 1 \pmod{p-1}$. Its scaling factor is

$$p^{-(k-1)/(p-1)} \prod_{i=1}^k p^{-(j_i-1)/(p-1)} = p^{-(N-1)/(p-1)}.$$

This proves (45) term by term. For the inverse, solve $f(g(x)) = x$ by degree. At an excluded degree N , all nonlinear terms built from earlier supported coefficients vanish. The nonzero linear coefficient of f then forces the inverse coefficient to vanish. $\square$

Definition 18.2. Let $\mathcal{C}_p$ consist of all $f(x) = \sum_{N \geq 1} a_N x^N \in \mathbb{Z}[[x]]$ supported in degrees $1 \pmod{p-1}$ such that

$$p^{w_p(N)} \mid a_N \quad \text{for every supported degree } N. \quad (46)$$

Write $\mathcal{C}_p^1$ for the members whose linear coefficient is one.

Theorem 18.3 (Closure and weighted seeds). *The class $\mathcal{C}_p$ is closed under addition and composition, and $\mathcal{C}_p^1$ is a group under composition. Every weighted seed*

$$W(x) = x + \sum_{j \geq 1} b_j x^{p^j}, \quad b_j \in \mathbb{Z}, \quad (47)$$

belongs to $\mathcal{C}_p^1$. Consequently every finite composition of such seeds and their inverses satisfies (46).

Proof. For an integral ordinary series $f \in \mathcal{S}_p$, Legendre's formula shows that

$$\begin{aligned} N![x^N]\mathcal{T}_p f \in \mathbb{Z} &\iff v_p(a_N) + v_p(N!) \geq d_p(N) \\ &\iff v_p(a_N) \geq w_p(N). \end{aligned} \quad (48)$$

The first equivalence uses the fact that the only denominator in $N!a_N/p^{d_p(N)}$ is a power of p . Thus $f \in \mathcal{C}_p$ exactly when $\mathcal{T}_p f$ is integral Hurwitz.

Addition is immediate. Composition follows from Lemmas 17.3 and 18.1, together with integrality of ordinary composition. If $f \in \mathcal{C}_p^1$, its ordinary inverse has integer coefficients by the triangular inverse recursion. Its scaled inverse is the Hurwitz inverse of $\mathcal{T}_p f$, so it lies in $\mathcal{C}_p^1$ as well. Finally, a p -power degree has digit sum one, hence weight zero; all weighted seeds satisfy the condition. $\square$

Proof of Theorem 16.1. Take every weighted seed in Theorem 18.3 to be F_p . Equivalence (48) gives the integer quotient in (39), and subtracting $v_p(N!)$ gives the valuation bound. If a coefficient is nonzero modulo p^q , its weight is at most $q - 1$, which is exactly (40). $\square$

[Write note, batch 114, 6 October 2026: second route to Theorem 1.2, and the answer to Research question 13.2.] At $p = 2$, (41) gives $2^{s_2(N)-1} \mid A_2(m, N)$ for every $m \in \mathbb{Z}$ and every $N \geq 1$. If $s_2(N) > q$, then $s_2(N) - 1 \geq q$ and so $2^q \mid A_2(m, N)$: this is the general theorem asked for in Research question 13.2, for every $q \geq 1$, for every $m \geq 1$ as asked, and also for $m \leq 0$. With $q = 2$ it is the first assertion of Theorem 1.2. If N is not a power of two, then $s_2(N) \geq 2$ and $A_2(m, N)$ is even, which is the second. Part I proves that theorem modulo 4 through balanced pairs and the carry transform (Lemma 3.1 and Proposition 4.2); the proof above scales F to an integral Hurwitz series and works at every precision at once. Neither proof uses the other. The actual residues at binary weight at most q , which the question's last sentence also mentions, are not determined here (Research question 27.1).

18.1 A small binary example

The scaling of the binary seed is

$$\mathcal{T}_2 F_2 = x + \frac{x^2}{2} + \frac{x^4}{8} + \frac{x^8}{128} + \cdots$$

Its exponential coefficients in degrees 1, 2, 4, 8 are 1, 1, 3, 315. Thus the apparent denominators disappear in exponential coordinates. The composition formula propagates that integrality through every iterate. At $N = 7$, it forces $v_2(A_2(m, 7)) \geq 2$ for every m . For $m = 2$ the coefficient is 8, so additional divisibility occurs. The theorem gives a lower bound, not an equality criterion.

18.2 An elementary combinatorial model

For $m \geq 0$, $A_p(m, N)$ counts rooted ordered trees with N leaves, all leaves at depth exactly m , and each internal vertex having p^j children for some $j \geq 0$. Unary vertices are allowed. For $m = 0$ there is one tree with one leaf. At the root, choosing p^j ordered subtrees gives the ordinary product $G(x)^{p^j}$, proving the model by induction on m . The model is the standard interpretation of ordinary-series substitution with specified arities.

There is also a labeled interpretation of the scaled integer

$$D_p(m, N) = \frac{N!}{p^{d_p(N)}} A_p(m, N).$$

At a vertex with $k = p^j$ child blocks, give its unordered set of children one of $k!/p^{(k-1)/(p-1)}$ local decorations. This number is an integer by Legendre. Repeated set-partition composition counts such decorated hierarchies on N labels and gives $D_p(m, N)$. The exponent in the scaling is consistent with every tree because

$$\sum_{v \text{ internal}} (\text{arity}(v) - 1) = N - 1. \quad (49)$$

Indeed, the sum of the arities counts all edges, while the number of edges is the number of vertices minus one. This explains the linear weight $d_p(N)$ independently of the digit notation.

19 Consequences for prime powers and OEIS diagonals

For a fixed modulus, the candidate exponents occupy only a few digit layers. The congruence class must be retained as well as the cutoff.

Prime	Modulus	Permitted positive digit sums
2	2, 4, 8, 16	respectively 1; 1, 2; 1, 2, 3; 1, 2, 3, 4
3	3, 9, 27	respectively 1; 1, 3; 1, 3, 5
5	5, 25, 125	respectively 1; 1, 5; 1, 5, 9

Membership in these lists is necessary, not sufficient, for a nonzero coefficient modulo the indicated modulus.

Proposition 19.1 (The odd-prime correction). *For every prime p , (37) holds. In particular, for every odd prime there is a nonzero coefficient modulo p^2 at digit sum $p > 2$.*

Proof. An outer power at least p^2 has minimum degree $p^2 > 2p - 1$. The outer degree-one term has no monomial of degree $2p - 1$. In the outer degree- p term, the only possible choice is one inner x^p and $p - 1$ inner x terms. There are exactly p choices. $\square$

[Write note, batch 114, 6 October 2026.] This is the counterexample to the suggestion in Part I's Research question 13.3, which is re-scoped there. At $p = 2$ the exponent 3 has binary weight two, in accordance with Part I.

19.1 The entire first residue layer modulo p

Over $\mathbb{F}_p$, a series supported at p -powers is additive. Let τ denote the Frobenius substitution $H(x) \mapsto H(x^p)$ on additive series. Composition of additive series with coefficients in $\mathbb{F}_p$ corresponds to multiplication of their formal series in τ . Consequently

$$F_p \longleftrightarrow 1 + \tau + \tau^2 + \cdots = (1 - \tau)^{-1}. \quad (50)$$

Formal expansion in τ gives, for every $m \in \mathbb{Z}$ and $r \geq 0$,

$$A_p(m, p^r) \equiv (-1)^r \binom{-m}{r} \pmod{p}. \quad (51)$$

For $m \geq 1$, this is the familiar expression $\binom{m+r-1}{r}$. Using $(-1)^r \binom{-m}{r}$ also handles $m = 0$ and negative iterates without a convention problem.

[Write note, batch 114, 6 October 2026: second route to Proposition 5.1.] Over $\mathbb{F}_2$, $P^2 = P(x^2)$ for every $P \in \mathbb{F}_2[[x]]$, so Part I's Frobenius T is τ , and $(1 - \tau)^{-1} = (1 + \tau)^{-1}$. Reduction modulo 2 commutes with composition, and a composition of additive series is additive, so the parity series P_m of (13) is supported at powers of two; for $m \geq 1$, (51) gives its coefficients $\binom{m+r-1}{r}$. This is (15). The mechanism is Part I's own; Part II adds every prime and every $m \in \mathbb{Z}$. The Lucas form $P_m = P_{S_m}$ of Proposition 5.1 is not restated here.

Corollary 19.2 (Every noninitial diagonal term is divisible by p). *Let $b_p(N) = A_p(N, N)$. Then $p \mid b_p(N)$ for every $N > 1$. At supported degrees $N > 1$,*

$$v_p(b_p(N)) \geq \max\{1, w_p(N)\}. \quad (52)$$

Proof. If the digit sum exceeds one, the support congruence forces it to be at least p , so Theorem 16.1 applies. It remains to treat $N = p^a$, $a \geq 1$. In $\mathbb{F}_p[[\tau]]$,

$$(1 - \tau)^{-p^a} = (1 - \tau^{p^a})^{-1}.$$

The coefficient of τ^a is zero because $0 < a < p^a$. Under (50), this is exactly $A_p(p^a, p^a)$ modulo p . $\square$

The binary instance of this corollary is already part of the previous repository report. The all-prime statement is included to show how the digit bound interacts with the moving iteration index. It does not turn the cutoff into a complete description of the diagonal.

[Write note, batch 114, 6 October 2026: second route to Proposition 6.1.] At $p = 2$ the corollary says that $a(n)$ is even for $n > 1$. When n is not a power of two, both proofs use that coefficients off the powers of two are even (Part I through Theorem 5.2(b), the proof above through Theorem 16.1). For $n = 2^a$, Part I applies Lucas's theorem to $\binom{2^a+a-1}{a}$; the proof above reads the same number modulo 2 as the coefficient of τ^a in $(1 - \tau)^{-2^a} = (1 - \tau^{2^a})^{-1}$ over $\mathbb{F}_2$, which vanishes because $0 < a < 2^a$.

20 The first possible residue of the second iterate

The divisibility theorem identifies where a coefficient can begin in its p -adic expansion. We next determine that first coefficient explicitly when the iterate is two. The proof is a direct multinomial argument, independent of the Hurwitz-series proof.

For $i \geq 0$, put

$$S_i(z) = z + z^p + \cdots + z^{p^i}. \quad (53)$$

For a polynomial P over $\mathbb{F}_p$, define the linear functional

$$\mathcal{L}_p(P) = \sum_{k \geq 0} [z^{p^k}]P. \quad (54)$$

The sum is finite. This functional is distinct from the Cartier operators introduced in Section 23.

Theorem 20.1 (Digit-polynomial residue formula). *Suppose $N \geq 1$, $N \equiv 1 \pmod{p-1}$, and $N = \sum_i d_i p^i$, $0 \leq d_i < p$. Write $w = w_p(N)$. Then*

$$\boxed{\frac{A_p(2, N)}{p^w} \equiv \frac{(-1)^w}{\prod_i d_i!} \mathcal{L}_p \left(\prod_i S_i(z)^{d_i} \right) \pmod{p}.} \quad (55)$$

Every denominator on the right is invertible in $\mathbb{F}_p$.

Lemma 20.2 (The factorial unit). *For $b = \sum_i b_i p^i \geq 0$, let $U_p(b) = b! / p^{v_p(b!)}$. Then*

$$U_p(b) \equiv (-1)^{v_p(b!)} \prod_i b_i! \pmod{p}. \quad (56)$$

Proof. Separate multiples of p from the other factors of $b!$. If $b = pc + r$, $0 \leq r < p$, their p -free product is $(-1)^c r!$ modulo p , by Wilson's theorem. The unit contribution of the multiples of p is $U_p(c)$. Iterating $U_p(b) \equiv (-1)^c r! U_p(c)$ gives the product of digit factorials and the sign exponent $\sum_{j \geq 1} \lfloor b/p^j \rfloor$. This is $v_p(b!)$. At $p = 2$ the sign is simply one. $\square$

Proof of Theorem 20.1. Expand the outer powers:

$$A_p(2, N) = \sum_{k \geq 0} \sum_{\substack{\sum_j m_j = p^k \\ \sum_j m_j p^j = N}} \frac{(p^k)!}{\prod_j m_j!}. \quad (57)$$

All sums are finite at fixed N . Legendre's formula gives each multinomial coefficient the valuation

$$\frac{\sum_j s_p(m_j) - 1}{p-1} \geq \frac{s_p(N) - 1}{p-1} = w. \quad (58)$$

Equality holds exactly when the sum $N = \sum_j m_j p^j$ has no carries. Writing $m_j = \sum_\ell e_{j,\ell} p^\ell$, this says

$$\sum_{j+\ell=i} e_{j,\ell} = d_i \quad (i \geq 0). \quad (59)$$

For a term of exact valuation w , Lemma 20.2 yields

$$p^{-w} \frac{(p^k)!}{\prod_j m_j!} \equiv \frac{(-1)^w}{\prod_{j,\ell} e_{j,\ell}!} \pmod{p}. \quad (60)$$

Here the digit factorial of p^k is one, and the difference of the factorial valuation exponents is exactly w .

In the expansion of $\prod_i S_i(z)^{d_i}$, an allocation (59) chooses z^{p^ℓ} for each digit unit assigned to (j, ℓ) , with $j + \ell = i$. Its coefficient is

$$\frac{\prod_i d_i!}{\prod_{j,\ell} e_{j,\ell}!},$$

and its exponent is $\sum_{j,\ell} e_{j,\ell} p^\ell = \sum_j m_j$. Thus $\mathcal{L}_p$ selects exactly the allocations satisfying the remaining constraint $\sum_j m_j = p^k$. Terms of larger valuation vanish after division by p^w and reduction modulo p . Summing (60) proves (55). $\square$

20.1 Examples and an efficient sparse calculation

At a pure power $N = p^r$, the formula gives $A_p(2, p^r) \equiv r + 1 \pmod{p}$, agreeing with (51). For $p = 2, N = 7$, the digit polynomial is

$$z(z + z^2)(z + z^2 + z^4).$$

Its pure-power coefficients sum to zero modulo two. Hence $A_2(2, 7)/4$ is even, explaining why the actual coefficient 8 has more divisibility than the lower bound forces.

Suppose $N < p^L$ and its digit sum is s . Expanding the digit polynomial one digit unit at a time requires at most L^s choices. A sparse dictionary can collect equal exponents modulo p without ever computing $A_p(2, N)$. For fixed p and fixed weight w , s is fixed, so this uses polynomially many operations in $L = O(\log N)$. Exponents have $O(\log N)$ bits. This is an algorithm for the first normalized residue, not for the entire large integer coefficient. The delivered `code/leading_residue.py` implements this formula.

20.2 Moving the highest nonzero digit

Theorem 20.3 (Highest-digit stability). *Let $1 \leq c < p$, $0 \leq R < p^r$, and suppose $c + s_p(R) > 1$ and $c + s_p(R) \equiv 1 \pmod{p-1}$. Set*

$$N_t = cp^{r+t} + R, \quad w = \frac{c + s_p(R) - 1}{p - 1}.$$

For every integer $t \geq 0$,

$$\frac{A_p(2, N_t)}{p^w} \equiv \frac{A_p(2, N_0)}{p^w} \pmod{p}. \quad (61)$$

Proof. It is enough to move the digit one place. Let P_R be the digit polynomial of R . Since $S_{r+1} = S_r + z^{p^{r+1}}$, the difference of the two relevant polynomials is

$$P_R(S_{r+1}^c - S_r^c) = \sum_{a=1}^c \binom{c}{a} P_R S_r^{c-a} z^{ap^{r+1}}.$$

Every monomial in the a th summand has exponent at most

$$ap^{r+1} + R + (c - a)p^r < p^{r+2}.$$

It is strictly greater than p^{r+1} : equality could occur only with $a = c = 1$ and $R = 0$, excluded by the digit-sum assumption. There is therefore no pure p -power exponent in the difference. The functional $\mathcal{L}_p$ gives equal values, while the factorial prefactor in (55) is unchanged. $\square$

The exception for digit sum one is necessary: the residue at p^r is $r + 1$ and changes when that digit moves.

20.3 Three infinite families with exact valuation

Theorem 20.4 (Sharp families for the second iterate). *For every prime p ,*

$$A_p(2, p^r + p - 1) \equiv p \pmod{p^2} \quad (r \geq 1), \quad (62)$$

$$A_p(2, p^r + (p - 1)(p + p^2)) \equiv p^2 \pmod{p^3} \quad (r \geq 3), \quad (63)$$

$$A_p(2, p^r + (p - 1)(p + p^2 + p^3)) \equiv p^3 \pmod{p^4} \quad (r \geq 4). \quad (64)$$

Thus the bound is attained at weights one, two and three in infinitely many degrees of the actual full lacunary second iterate.

Proof. For (62), the digit polynomial is $z^{p-1}S_r$. Its only pure-power term is z^p , with coefficient one. Wilson gives $-1/(p-1)! \equiv 1 \pmod{p}$.

For (63), highest-digit stability reduces to $r = 3$. Terms containing the highest monomial z^{p^3} in $S_1^{p-1}S_2^{p-1}S_3$ lie strictly between p^3 and p^4 . Consequently, over $\mathbb{F}_p$,

$$\mathcal{L}_p(S_1^{p-1}S_2^{p-1}S_3) = \mathcal{L}_p(S_1^{p-1}S_2^p) = 1.$$

Indeed, $S_2^p = z^p + z^{p^2} + z^{p^3}$, and the sole contribution is the leading term of S_1^{p-1} , of degree $p^2 - p$, paired with z^p . The prefactor is $1/((p-1)!)^2 \equiv 1$.

For (64), reduce to $r = 4$ and put $T = S_1^{p-1}S_2^{p-1}$. Removing the highest monomial of S_4 and using Frobenius gives

$$\mathcal{L}_p(TS_3^{p-1}S_4) = \mathcal{L}_p(TS_3^p).$$

The degree of T is $p^3 - p$. The only contributions are

$$[z^{p^2-p}]T + [z^{p^3-p}]T + [z^{p^3-p^2}]T = 1 + 1 - 1 = 1 \quad \text{in } \mathbb{F}_p. \quad (65)$$

For completeness, the middle coefficient is the leading coefficient. The first is the coefficient of z^{p^2-p} in S_1^{2p-2} , namely $\binom{2p-2}{p-2} \equiv 1$. For the third, expand

$$T = \sum_{a=0}^{p-1} \binom{p-1}{a} z^{ap^2} S_1^{2p-2-a}.$$

At degree $(p-1)p^2$, only $a = p-2$ contributes. Its coefficient is $\binom{p-1}{p-2} [z^{p^2}]S_1^p = -1$. If $a \leq p-3$, the required degree exceeds the degree of the remaining factor; if $a = p-1$, it would require a constant term. These arguments also hold at $p = 2$. Finally, $(-1)^3/((p-1)!)^3 \equiv 1$, which proves the third congruence. $\square$

For $p = 2$, the three families are $2^r + 1$, $2^r + 6$ and $2^r + 14$, with exact valuations one, two and three, respectively. They do not assert that every exponent of the same binary weight is sharp.

21 Optimality in the larger class

The preceding families concern the particular full seed F_p . There is a stronger, exactly solvable sharpness example for $\mathcal{C}_p^1$. It comes from a classical Fuss–Catalan inverse, rather than a new enumeration of that family.

Theorem 21.1 (An exact sharpness certificate). *Let $B_p(x) = x + x^p$ and $C_p(x) = B_p^{\circ(-1)}(x)$. For $k \geq 0$ and $N = 1 + (p-1)k$,*

$$[x^N]C_p(x) = \frac{(-1)^k}{N} \binom{pk}{k}, \quad v_p([x^N]C_p) = w_p(N). \quad (66)$$

All other coefficients vanish. Thus the digit valuation bound is optimal at every supported degree in the class $\mathcal{C}_p^1$.

Proof. Formal Lagrange inversion [9] for $x = u(1 + u^{p-1})$ gives

$$[x^N]C_p(x) = \frac{1}{N} [u^{N-1}](1 + u^{p-1})^{-N}.$$

For $N - 1 = (p - 1)k$, this equals $(-1)^k \binom{N+k-1}{k} / N$, proving the first formula. One quick formal justification of this use of Lagrange is to substitute $x = u\phi(u)$ in the residue $\text{Res}_x C_p(x)x^{-N-1} dx$. Integration by parts reduces it to $N^{-1} \text{Res}_u u^{-N} \phi(u)^{-N} du$. No analytic integration is involved.

The coefficient is an integer because B_p is an integral series tangent to the identity. Legendre and $s_p(pk) = s_p(k)$ give

$$v_p \binom{pk}{k} = \frac{s_p((p-1)k)}{p-1} = \frac{s_p(N-1)}{p-1}.$$

Subtracting one from N changes its digit sum according to

$$s_p(N-1) = s_p(N) - 1 + (p-1)v_p(N).$$

Subtract $v_p(N)$ from the binomial valuation to obtain the second identity in (66). $\square$

For every $q \geq 1$, the exponent $N = p^q - (p-1)$ has $s_p(N) = 1 + (q-1)(p-1)$ and inverse coefficient of valuation exactly $q-1$. Thus the cutoff in (40) cannot be lowered in the larger class. At $p=2$, (66) is the usual signed Catalan inverse and its familiar binary valuation.

There is even existential sharpness using positive iterates of the polynomial B_p . Fix a degree cutoff and a modulus p^q . The truncated series $x + O(x^2)$ over $\mathbb{Z}/p^q\mathbb{Z}$ form a finite group under composition, so the inverse of B_p is some positive power of B_p in that group. This observation gives no small useful iteration index and says nothing comparable about a positive power of the special full seed F_p .

22 Sparse support and typical divisibility

The uniformity in the iteration number has counting consequences even when that number varies arbitrarily with the exponent.

Theorem 22.1 (An exact candidate count). *Fix $p, q \geq 1$, with p prime, and put $R = 1 + (q-1)(p-1)$. The number of positive integers $N < p^L$ permitted by (40) is*

$$C_{p,q}(L) = \sum_{j=0}^{q-1} [u^{1+j(p-1)}] (1 + u + \cdots + u^{p-1})^L. \quad (67)$$

For fixed p, q ,

$$C_{p,q}(L) = \frac{L^R}{R!} + O_{p,q}(L^{R-1}). \quad (68)$$

For any map $m : \mathbb{N}_{\geq 1} \rightarrow \mathbb{Z}$, the number of indices $N < p^L$ at which $A_p(m(N), N) \not\equiv 0 \pmod{p^q}$ is at most $C_{p,q}(L)$. In particular this support has natural density zero.

Proof. The L digits of an integer in $[0, p^L)$ are independently chosen from $0, \dots, p-1$. The polynomial in (67) enumerates their sum. The congruence $N \equiv s_p(N) \pmod{p-1}$ gives exactly the displayed layers.

For fixed digit sum s , the choices using s distinct positions, each with digit one, number $\binom{L}{s} = L^s/s! + O_s(L^{s-1})$. Every other choice uses at most $s-1$ nonzero positions and contributes $O_{p,s}(L^{s-1})$. The largest permitted sum is R , which gives (68). The final assertions follow from Theorem 16.1; passing from powers of p to arbitrary cutoffs changes only a bounded multiplicative factor. $\square$

For the binary sequences, the exact bound simplifies to

$$C_{2,q}(L) = \sum_{j=1}^q \binom{L}{j}. \quad (69)$$

For example, below 2^{20} there are at most 1350 candidate indices modulo 8, and at most 6195 modulo 16, for every choice of $m(N)$. Actual support can be substantially smaller. The earlier report's linear-in- L count for the exceptional binary diagonal modulo four is finer than our quadratic candidate bound at $q = 2$; we do not replace that result.

[Write note, batch 114, 6 October 2026.] The earlier report's count is Theorem 9.2.

Theorem 22.2 (A quantitative density-one lower bound). *Fix $0 < \alpha < 1/2$. There are constants $c = c(p, \alpha) > 0$ and $K = K(p, \alpha) > 0$ such that, for every $L \geq 1$ and every choice of integer iteration indices $m(N)$,*

$$\#\{1 \leq N < p^L : v_p(A_p(m(N), N)) < \alpha L\} \leq K p^L e^{-cL}. \quad (70)$$

Consequently, for every $\varepsilon > 0$, on a set of natural density one,

$$v_p(A_p(m(N), N)) \geq (1/2 - \varepsilon) \log_p N. \quad (71)$$

This is a lower bound; no typical equality is asserted.

Proof. If the valuation is less than αL , the coefficient is nonzero and Theorem 16.1 forces $s_p(N) < (p-1)\alpha L + 1$. Let $X_1, \dots, X_L$ be independent uniform digits in $\{0, \dots, p-1\}$. For any $t > 0$, Markov's inequality applied to $e^{-t} \sum X_i$ gives

$$\Pr\left(\sum_i X_i < (p-1)\alpha L + 1\right) \leq e^t \left(e^{t(p-1)\alpha} \frac{1 + e^{-t} + \dots + e^{-(p-1)t}}{p} \right)^L.$$

The logarithm of the parenthesized quantity vanishes at $t = 0$ and has derivative $(p-1)(\alpha - 1/2) < 0$ there. A sufficiently small fixed $t > 0$ makes it e^{-c} with $c > 0$. This proves (70), with $K = e^t$.

Choose α strictly between $1/2 - \varepsilon$ and $1/2$. For $p^{L-1} \leq N < p^L$, failure of (71) implies a valuation below αL . The estimate on geometric cutoffs therefore makes the relative number of failures tend to zero at all cutoffs. $\square$

23 Finite automata for each fixed iterate

The sparse-support theorem does not itself provide all surviving residues. A different argument gives a finite computation for every coefficient modulo a fixed prime power, once the iterate is fixed. We give the construction explicitly; it does not require the characteristic-zero generating function to be algebraic or a diagonal of a rational function.

Definition 23.1. For $0 \leq r < p$, the Cartier operator on $R[[x]]$ is

$$\Lambda_r \left(\sum_{n \geq 0} a_n x^n \right) = \sum_{n \geq 0} a_{pn+r} x^n. \quad (72)$$

A sequence with values in a finite set is p -automatic if a finite automaton with output reads the base- p digits of its index and returns the corresponding value.

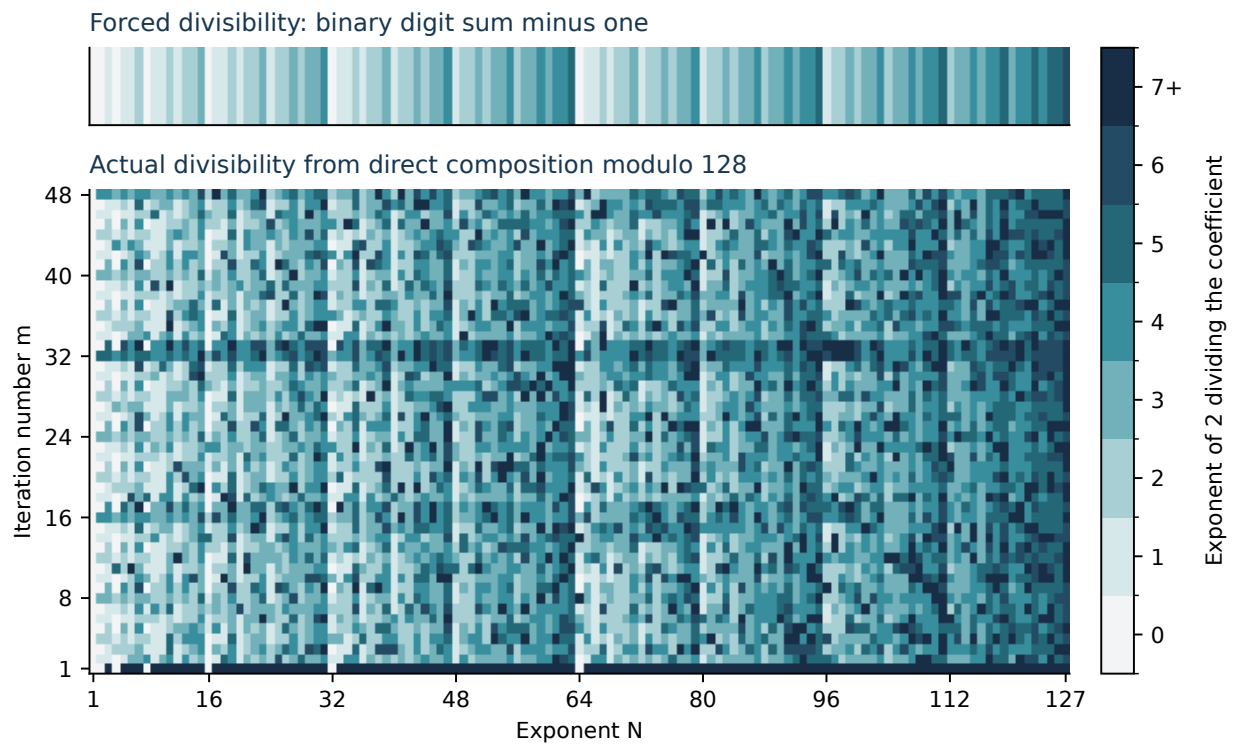


Figure 1: Exact binary divisibility in degrees $1 \leq N \leq 127$. The upper strip is the uniform lower bound $s_2(N) - 1$. The lower panel shows $v_2(A_2(m, N))$ for $1 \leq m \leq 48$, capped at seven. A value seven means divisibility by 128; it includes exact zero coefficients. The data come from direct ordinary-series composition modulo 128, without digit-based pruning. The theorem explains the common forced layers, while the additional colors record cancellations beyond the bound.

We use the least-significant-digit reading convention. The usual equivalent characterization is finiteness of the p -kernel [13, Theorem 1.8]. In the proof below, the automaton is built directly from a finite Cartier-stable module.

Lemma 23.2 (Lifting Frobenius). *For $H(x) \in \mathbb{Z}[[x]]$ and $j \geq 1$,*

$$H(x)^{p^j} \equiv H(x^p)^{p^{j-1}} \pmod{p^j}. \quad (73)$$

Consequently, if $H(0) = 0$ and $q \geq 1$,

$$F_p(H(x)) \equiv \sum_{j=0}^{q-2} H(x)^{p^j} + \sum_{k \geq 0} \left(H(x)^{p^{q-1}} \right) (x^{p^k}) \pmod{p^q}, \quad (74)$$

where the first sum is empty for $q = 1$.

Proof. The usual Frobenius congruence gives $H(x)^p \equiv H(x^p) \pmod{p}$. If $U \equiv V \pmod{p^a}$ with $a \geq 1$, the binomial expansion shows $U^p \equiv V^p \pmod{p^{a+1}}$, including $p = 2$. Applying this $j - 1$ times proves (73), the congruence also used in the proof of [13, Proposition 1.9].

For $j \geq q - 1$, telescope (73) at successive substitutions and powers to obtain

$$H(x)^{p^j} \equiv H(x^{p^{j-q+1}})^{p^{q-1}} \pmod{p^q}.$$

At $j = q - 1$ this is an identity; for larger j each discarded difference is divisible by at least p^q . Summing yields (74). The zero constant term makes all infinite sums coefficientwise finite. $\square$

Lemma 23.3 (Two finite-module constructions). *Let R be a finite commutative ring.*

- (i) *For every integer $k \geq 1$, if a module with d generators $u_1, \dots, u_d$ is stable under every Λ_r , then the module generated by*

$$x^c u_1^{\alpha_1} \cdots u_d^{\alpha_d}, \quad 0 \leq c \leq k - 1, \quad \sum_i \alpha_i = k, \quad (75)$$

is Cartier-stable and contains H^k for every H in the original module. It has at most $k \binom{d+k-1}{k}$ generators.

- (ii) *If $B(0) = 0$ belongs to a finitely generated Cartier-stable module, then adjoining*

$$\mathcal{S}(B) = \sum_{j \geq 0} B(x^{p^j}) \quad (76)$$

gives another Cartier-stable module.

Proof. Every series satisfies $u(x) = \sum_{a=0}^{p-1} x^a (\Lambda_a u)(x^p)$. Apply this identity to the k factors in a generator (75). A selected term has exponent $c + a_1 + \cdots + a_k$. Under Λ_r , it contributes only when that exponent is r modulo p , and its remaining power of x is

$$\frac{c + a_1 + \cdots + a_k - r}{p} \in \{0, \dots, k - 1\}.$$

Each transformed factor is an R -linear combination of the u_i . The result is therefore in the stated module. The count is the number of degree- k commutative monomials times the k shifts.

For part (ii), put $C = \mathcal{S}(B)$. The identity $C(x) = B(x) + C(x^p)$ gives

$$\Lambda_0 C = \Lambda_0 B + C, \quad \Lambda_r C = \Lambda_r B \quad (1 \leq r < p). \quad (77)$$

Thus one additional generator suffices. $\square$

Theorem 23.4 (Constructive automaticity). *For every fixed prime p , precision $q \geq 1$, and iterate $m \geq 0$, the sequence*

$$(A_p(m, N) \bmod p^q)_{N \geq 0} \quad (78)$$

is p -automatic. For $m \geq 1$, a Cartier representation with d_m generators can be constructed with $d_1 = 2$ and

$$d_{m+1} \leq 1 + \sum_{j=0}^{q-1} p^j \binom{d_m + p^j - 1}{p^j}. \quad (79)$$

Such a representation gives an automaton with at most p^{qd_m} states and allows a coefficient query using $O(d_m^2 \log_p(N+1))$ ring operations after setup.

Proof. Work over $R = \mathbb{Z}/p^q\mathbb{Z}$. The module spanned by $1, F_p$ is stable:

$$\Lambda_0 F_p = F_p, \quad \Lambda_1 F_p = 1, \quad \Lambda_r F_p = 0 \quad (r \geq 2).$$

For the identity iterate, use the stable span of $1, x$ instead. Starting from the representation of $F_p^{\circ m}$, apply Lemma 23.3(i) to each of the powers $1, p, \dots, p^{q-1}$ in (74). Take the direct sum of those modules and adjoin the one Mahler-sum generator from part (ii). This represents $F_p^{\circ(m+1)}$ and proves (79).

A module generated by d_m elements over R has at most $|R|^{d_m} = p^{qd_m}$ elements. Its Cartier orbit is therefore finite. Choose explicit transition matrices for the generators. Starting at the vector representing $F_p^{\circ m}$, apply the matrix for each successive least-significant digit of N . The constant coefficient of the resulting series is $A_p(m, N)$. This is a finite automaton with output. Dense matrix-vector multiplication gives the stated operation bound; sparse matrices are often much smaller. $\square$

The construction is finite but its setup bound grows quickly with m and q . It is not a uniformly inexpensive algorithm in all three parameters. Nor does substituting $m = N$ in the theorem produce one finite automaton for the diagonal: the representation would change with the input index.

The supplied implementation constructs sparse transition matrices, without enumerating every automaton state. Four exact cross-checks give the following unreduced representation sizes:

p	q	m	Generators	Nonzero transition entries
2	2	2	9	19
2	3	2	29	81
3	2	2	15	38
2	2	3	100	371

These are sizes of the displayed construction, not minimal state counts. They are independently verified through degree 512 in `data/cartier_verification.json`.

24 p -adic iteration and periods in the iteration parameter

The main divisibility result also controls iteration at a p -adic time. The construction below is classical finite-difference interpolation applied coefficientwise; its useful additional feature here is the digit-dependent divisibility of every finite-difference coefficient.

Let C be the linear substitution operator $C(H) = H \circ F_p$ on $\mathbb{Q}[[x]]$, and put $D = C - I$. Since $F_p = x + O(x^p)$,

$$\text{ord}_x(DH) \geq \text{ord}_x(H) + p - 1 \quad \text{when } H \in x\mathbb{Q}[[x]], \quad DH \neq 0. \quad (80)$$

This follows first for a monomial from $F_p^j - x^j \in x^{j+p-1}\mathbb{Z}[[x]]$, then by linearity.

Theorem 24.1 (A coefficientwise continuous iteration group). *For $z \in \mathbb{Z}_p$, define*

$$F_p^{\circ z}(x) = \sum_{k \geq 0} \binom{z}{k} D^k x. \quad (81)$$

Each coefficient is a finite sum, belongs to $\mathbb{Z}_p$, and depends continuously on z . The definition agrees with ordinary integer iteration and satisfies

$$F_p^{\circ(z+t)} = F_p^{\circ z} \circ F_p^{\circ t}, \quad F_p^{\circ 0} = x, \quad [x^p]F_p^{\circ z} = z. \quad (82)$$

At a supported degree $N > 1$, its coefficient has the form

$$P_N(z) = \sum_{k=0}^{d_p(N)} b_{N,k} \binom{z}{k}, \quad b_{N,k} \in p^{w_p(N)}\mathbb{Z}. \quad (83)$$

Thus $P_N(z) \in p^{w_p(N)}\mathbb{Z}_p$ for every $z \in \mathbb{Z}_p$.

Proof. The order increase (80) proves coefficientwise finiteness and the degree bound in (83). For a nonnegative integer m , the finite binomial identity $(I + D)^m = C^m$ gives ordinary iteration.

At degree $N > 1$,

$$b_{N,k} = [x^N]D^k x = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} A_p(j, N). \quad (84)$$

Every term is divisible by $p^{w_p(N)}$ by Theorem 16.1; the identity-iterate term is zero. This proves the asserted divisibility of the finite differences.

The polynomial $\binom{z}{k}$ maps $\mathbb{Z}_p$ into $\mathbb{Z}_p$: it is continuous over $\mathbb{Q}_p$, and its integer values on the dense set of nonnegative integers are p -integral. Hence every coefficient in (81) is continuous and p -integral. For each fixed degree, composition uses finitely many coefficient polynomials in z, t . The group identity holds on pairs of nonnegative integers and extends by continuity to $\mathbb{Z}_p^2$. It identifies negative integers with formal inverse iteration. Finally, Dx has coefficient one at x^p , and $D^k x$ has no such term for $k \geq 2$, proving the last identity. $\square$

The continuity here is in the *coefficientwise p -adic* topology, not the pure x -adic topology. For example, a small nonzero p -adic parameter still produces the term zx^p . The parameterization is injective because that coefficient recovers z . Also, $P_N(z)$ is an integer-valued rational polynomial; it need not be a polynomial in $\mathbb{Z}[z]$.

Theorem 24.2 (A digit-sensitive sufficient period). *Let $N > 1$ be supported, $d = d_p(N)$, $w = w_p(N)$, and $q \geq 1$. If $q \leq w$, then P_N is identically zero modulo p^q . If $q > w$, its value modulo p^q depends only on*

$$z \pmod{p^h}, \quad h = q - w + \lfloor \log_p d \rfloor. \quad (85)$$

No minimality of this period is asserted.

Proof. Only the case $q > w$ needs proof. Then $p^h > d$. For $1 \leq i \leq d$,

$$v_p \binom{p^h}{i} = h - v_p(i) \geq q - w. \quad (86)$$

Indeed, $\binom{p^h}{i} = (p^h/i) \binom{p^h-1}{i-1}$, and the second factor is a p -adic unit, as is seen by reducing $(1+X)^{p^h-1}$ modulo p . Vandermonde's identity gives

$$\binom{z+p^h}{k} - \binom{z}{k} = \sum_{i=1}^k \binom{p^h}{i} \binom{z}{k-i}.$$

Multiplying by $b_{N,k} \in p^w \mathbb{Z}$ and summing proves $P_N(z+p^h) \equiv P_N(z) \pmod{p^q}$. Integer multiples follow by iteration; all p -adic multiples follow by continuity. $\square$

The degree-one coefficient is identically one and needs no period formula. Unsupported coefficients are identically zero. The period estimate can be far from minimal, particularly at a pure-power degree, where (51) already gives a smaller description modulo p .

25 Exact computations and reproducibility

Every unbounded assertion above rests on its written proof. The programs serve as independent checks of formulas, indexing, normalization and implementation. They use exact integer or finite-ring arithmetic; the plotting dependencies are matplotlib and NumPy.

25.1 Three independent computational routes

`code/verify.py` directly composes ordinary power series without discarding coefficients on the basis of the digit theorem. It checks support and divisibility for $p = 2, 3, 5, 7$, iterates $1 \leq m \leq 8$ and degrees through 128, and checks inverse series through degree 35. Separately, it implements the set-partition Bell recurrence and compares the scaled coefficients with ordinary composition through degree 28. It also checks the Fuss–Catalan valuation, the first ten terms of each of the three OEIS diagonals, the odd-prime examples, and sufficient periods.

`code/leading_residue.py` constructs the digit polynomial over $\mathbb{F}_p$ using a sparse dictionary. Its verifier compares against a separate ordinary-composition routine for all degrees 0 through 300 and $p = 2, 3, 5, 7$. It additionally checks highest-digit stability and the three sharp families.

`code/cartier.py` implements the finite-module construction from Section 23. Its verifier compares coefficient queries against direct Cauchy composition through degree 512 in four parameter settings. It can also export the sparse transition representation as JSON. It guards against excessive setup sizes; these guards are resource limits, not restrictions of the theorem.

Recorded independent check	Number
Coefficient support, valuation, and scaled integrality	4236
Ordinary composition versus Bell composition	336
Exact Fuss–Catalan valuations	750
OEIS initial-term fixtures	30
Digit formula versus ordinary second composition	1204
Highest-digit stability comparisons	70
Sharp-family evaluations	18
Cartier queries versus ordinary composition	2052

The counts above describe finite checks, not a count of proved theorems. Exact receipts are supplied in the `data/` directory. The heatmap uses another direct modular composition calculation. No numerical fit, floating-point valuation, or guessed recurrence is used in any proof.

25.2 Reproduction commands

From the package directory, run:

```
python3 code/verify.py --output data/verification.json \
    --heatmap data/valuation_grid.json
python3 code/leading_residue.py \
    --verify data/leading_residue_verification.json
python3 code/cartier.py --verify \
    --verification-output data/cartier_verification.json
python3 code/make_figure.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The first and second commands explicitly name their output files. The Cartier verifier's output path and query options are documented by its `-help` output and in `README.txt`. The supplied figure PDF permits compilation without matplotlib.

[Write note, batch 114, 6 October 2026.] These are the commands of the delivered layout. In this report the files carry the prefix `02-digitsum-` (Table 3); `make_figure.py` reads and writes fixed delivery paths under its parent directory, the other programs write where they are told, and on Windows they write CRLF line endings. Run them on a copy in the delivered layout, never in place; the README gives the route, and Section 15.7 reports the intake's rerun.

26 Source audit and limits of the conclusions

The inspected repository article is

```
SetTheory/Cardinals/docs/reports/congruences-and-valuations/
iterated-series/a168362-lacunary-iterates-mod4/article.tex.
```

Its Git blob is 46147fca8f8fb912f7132d504b21217c0c248844. That exact blob was verified at repository commit 3412ae0741b95b3fb6d7a43477b7fac7723c36d8. The snapshot was inspected on October 5, 2026. Its Questions 13.2 and 13.3 are the direct targets of this article.

The repository's report on compositional tree-series congruences studies different labeled-tree exponential generating functions. Its wider-class discussion does not include arbitrary ordered arity restrictions. A sibling on self-iterating series uses negative iterates and p -adic iteration in another functional equation. These are relevant methodological neighbors, not sources for the present lacunary digit bound. The iterated-Bell diagonal report contains an analytic method pointer for A168362, without asserting its ratio conjecture.

The external checks covered the current OEIS definitions and comments, the Hurwitz-series composition reference [12], and the prime-power Cartier and automaton framework [13]. Targeted searches for the exact sequence and for lacunary composition with digit-sum divisibility did not locate the present application. This is a bounded search result, not a proof of publication priority. The classical tools are derived in the article, so no unverified repository claim is a proof dependency.

The important limitations are mathematical and specific:

- The digit cutoff is a necessary support condition. Extra divisibility and cancellation are common. Except for the explicit second-iterate formula and families, no equality classification for the valuation is asserted.
- Universal optimality is proved in $\mathcal{C}_p^1$ through the inverse of $x + x^p$. For the particular full $F_p^{\circ 2}$, only the stated weights one, two and three are proved sharp in infinite families here.
- Automaticity is proved for each fixed nonnegative integer iterate and fixed precision. No automaticity claim is made for the diagonal $m = N$, for arbitrary p -adic time, or for negative iterates modulo higher prime powers.
- The period in Theorem 24.2 is sufficient and may be large. The setup costs of the automaton construction can also grow rapidly.
- The source report's real asymptotic ratio question is unresolved by these congruence arguments. The previous exact modulo-four diagonal classification and its counting constant are not re-proved or claimed as new.

[Write note, batch 114, 6 October 2026.] The pinned blob is Part I as printed here, without its notes of 6 October 2026 (Section 15.1). The statements above about the neighbouring reports were checked and are correct: `compositional-tree-series-congruences` says that its wider class of tree specifications does not include arbitrary degree restrictions or ordered children; `a396807-self-iterating-series` treats iterates of every integer order and p -adic-time iteration of its own series; and `a139383-iterated-bell-diagonals` names Part I's refined asymptotic as open and makes no claim about it. The “two independent mathematical reviews” of the source audit are, by the delivered `data/document_qa.json`, two AI audits; this is a process claim the intake cannot verify. No human review and no formal verification is claimed.

27 Further research questions and proposed approaches

The following questions arise from the results proved here. Their status refers to this article and the inspected sources, rather than an exhaustive claim about all literature.

[Write note, batch 114, 6 October 2026.] Under Vladimir Reshetnikov's standing rule of 4 October 2026, unproved claims are kept as open questions. These ten are the manuscript's, with its proposed approaches; the write found no other unproved claim in it and no statement of it to be wrong. Research question 27.9 restates Part I's Research question 13.1; Research question 27.1 continues Research question 13.2 on the diagonal, where the iteration index moves with N ; Research question 27.3 continues the re-scoped Research question 13.3.

Research question 27.1 (Exact higher-modulus diagonals). Classify $A_2(N, N) \bmod 8$ and $\bmod 16$, including the surviving three- and four-bit layers, and obtain the sharp count of nonzero indices below X .

The new filtration removes all larger-weight indices, but the iteration number is still N . A useful approach is to combine finite differences in the iteration parameter with a carry calculus restricted to the finitely many permitted digit layers. The earlier modulo-four diagonal classification gives a stringent specialization check.

Research question 27.2 (Arbitrary-depth sharpness for the full second iterate). For every p and every $w \geq 4$, does some degree N satisfy $v_p(A_p(2, N)) = w_p(N) = w$? Can one give uniform infinite families?

By highest-digit stability, one example at a given digit pattern immediately gives infinitely many. The remaining problem is to find a pattern whose digit-polynomial functional in (55) is nonzero. The Fuss–Catalan example does not answer this more specific question.

Research question 27.3 (Leading residues of higher iterates). Find a digit-level formula for $A_p(m, N)/p^{w_p(N)} \bmod p$ for general fixed m , comparable to Theorem 20.1.

Repeated Bell composition suggests a hierarchy of carry-free digit allocations. A successful formula should keep the number of states controlled in m and the digit weight, rather than enumerate all labeled set partitions of N .

Research question 27.4 (Minimal Cartier representations). Determine the minimal automaton or minimal linear representation for selected (p, q, m) , and improve (79).

The current construction deliberately keeps dependent generators. Finite-ring relations among its shifted monomials can make the actual representation much smaller. Over $\mathbb{Z}/p^q\mathbb{Z}$, reduction must respect nonunits; field Gaussian elimination alone is not a general minimization procedure.

Research question 27.5 (Inverse-iterate automaticity). Are the coefficient sequences of $F_p^{o(-m)}$ modulo p^q p -automatic for every fixed positive m, q ?

The valuation theorem and p -adic period theorem cover these iterates, while the finite forward-composition construction does not. An effective inverse closure theorem adapted to the Mahler-sum representation would connect the two parts of the paper.

Research question 27.6 (Optimal iteration periods). Determine the least period of $A_p(m, N) \bmod p^q$ as a function of m . Describe this period from the base- p digits of N .

The sufficient period comes from bounding all binomial terms by their maximum degree. The valuations and cancellations of the individual finite differences $b_{N,k}$ should often improve it. Pure-power degrees give accessible exact test cases.

Research question 27.7 (Typical excess valuation). For fixed m , or for $m = N$, study $v_p(A_p(m, N)) - w_p(N)$ at nonzero coefficients and determine its distribution over digit strings.

The current density-one theorem supplies only the forced part. For a fixed row the finite automaton can support exact frequency calculations at each modulus. Passing to all precisions or to a moving diagonal requires additional uniform control.

Research question 27.8 (Which weighted seeds retain automaticity?). For $W(x) = x + \sum b_j x^{p^j}$, characterize the coefficient sequences (b_j) for which fixed iterates of W are p -automatic modulo each p^q .

The divisibility theorem allows arbitrary integer weights. Automaticity is more restrictive: already the seed can encode an arbitrary sequence along the powers of p . Eventually periodic weights in the exponent index provide a natural first class with finite Mahler systems.

Research question 27.9 (The real-growth diagonal problem). Prove or refute the recorded conjecture $A_2(N+1, N+1)/A_2(N, N) \sim eN$, and determine a multiplicative asymptotic for the diagonal if it holds.

The ordered level-tree interpretation gives a concrete counting model. The present divisibility results impose arithmetic constraints but do not estimate the sizes of its positive coefficients. A saddle-point or branching analysis must also handle the fact that depth and leaf count grow together.

Research question 27.10 (Formal verification). Formalize the digit-divisibility theorem with the smallest reusable mathematical interface.

The proof separates naturally into factorial valuation, integral set-partition composition, and the weighted scaling identity. The core theorem needs no analytic estimates or automata library. A second stage could verify the digit-polynomial formula and finite transition certificates for concrete automata.

Appendices

[Write note, batch 114, 6 October 2026.] Appendices A and B belong to Part I; Appendices C and D belong to Part II.

A Proposed OEIS update text

The following is a concise draft, not submitted to OEIS.

Let $F(x) = \sum_{j \geq 0} x^{2^j}$ and $a(n) = [x^n]F^{\circ n}(x)$. For $s \geq 1$ put $M_s = 2^s - 1$ and define $R_s \subseteq \mathbb{Z}/2^s\mathbb{Z}$ by

$$\mathbf{1}_{d \in R_s} \equiv \mathbf{1}_{d=0} + \sum_{i=0}^s \mathbf{1}_{i \text{ OR } ((i+d) \bmod 2^s) = M_s} \pmod{2}.$$

Then $a(n)$ is even for every $n > 1$. Moreover, $a(n) \equiv 2 \pmod{4}$ exactly for $n \in \{2, 3, 4, 6\}$ and for $n = 2^r + 2^s$ with $r > s \geq 1$, $r \geq 3$, and $r - s \bmod 2^s \in R_s$; all other $n > 1$ give $a(n) \equiv 0 \pmod{4}$. The next exceptional indices after the currently listed 520 are 1028, 1032, 2056, 2064, 4100, 4104, $\dots$. If $E(X)$ counts the exceptions up to X , then $E(X) = \kappa \log_2 X + O((\log \log X)^3)$, where $\kappa = \sum_{s \geq 1} |R_s|/2^s = 2.323930059308106 \dots$

B Artifact inventory

[Write note, batch 7301.] In the collection, `article.tex` is the delivered source with prefixed labels and these notes, `article.pdf` is a build of it, and `README.md` is the collection README; the delivered PDF and README, a file list `MANIFEST.txt` and the checksum list `SHA256SUMS.txt` are not shipped. The other files below are shipped under the names shown, byte-identical to the delivery.

The accompanying archive contains:

File	Purpose
<code>article.tex</code> , <code>article.pdf</code>	manuscript source and compiled PDF
<code>code/verify_a168362.py</code>	exact standard-library verifier
<code>data/verification.json</code>	machine-readable PASS record
<code>data/verification_run.txt</code>	complete verifier stdout
<code>data/residue_sets.csv</code>	R_s for $1 \leq s \leq 64$
<code>data/exceptional_indices_to_10m.txt</code>	exceptions through 10^7
<code>oeis_update_draft.txt</code>	plain-text version of the proposed update
<code>README.md</code>	build and verification instructions

C A concise theorem suitable for an OEIS note

The following text is a draft for mathematical review, not a record of a submission.

[Write note, batch 114, 6 October 2026.] Nothing was submitted to OEIS. Part I's Appendix A is a draft of the same kind for the modulo-four theorem.

Let $F(x) = \sum_{j \geq 0} x^{2^j}$. For every integer m and $N \geq 1$, the coefficient $[x^N]F^{\circ m}(x)$ is divisible by $2^{s_2(N)-1}$, where $s_2(N)$ counts the ones in the binary expansion of N ; negative m denotes compositional inversion. In particular A168362, A168365 and A168366 are divisible by 2^q whenever $s_2(N) > q$. The assertion follows by observing that $\sum_{j \geq 0} x^{2^j}/2^{2^j-1}$ has integer exponential coefficients and that this property is preserved by composition and inversion. Legendre's formula then gives the stated exponent.

D Package inventory

The article is self-contained apart from its supplied figure. The ZIP contains the following principal files:

File	Purpose
<code>article.tex</code> , <code>article.pdf</code>	Complete source and compiled article
<code>README.txt</code>	Build instructions and example queries
<code>SOURCE_AUDIT.txt</code>	Source pin, provenance and novelty boundaries
<code>code/verify.py</code>	Ordinary and Bell composition; inverse, valuation and period checks
<code>code/leading_residue.py</code>	Exact sparse first-residue algorithm
<code>code/cartier.py</code>	Sparse finite-module construction and coefficient queries
<code>code/make_figure.py</code>	Recreates the scientific figure
<code>data/</code>	Exact verification receipts and figure data
<code>figures/valuation_grid.pdf</code>	Vector figure used by the article

[Write note, batch 114, 6 October 2026.] Of these, `article.tex`, `article.pdf` and `README.txt`, and the checksum list `SHA256SUMS.txt`, are not shipped; the others are shipped under the names of Table 3, with `data/document_qa.json` and `figures/valuation_grid.png`.

References

- [1] P. D. Hanna, *A168362: coefficient of x^n in the n -th iteration of $\sum_{k \geq 0} x^{2^k}$* , The On-Line Encyclopedia of Integer Sequences, <https://oeis.org/A168362>, accessed October 1, 2026.
- [2] P. D. Hanna, *A168365: coefficient of x^n in the $(n-1)$ -st iteration of $\sum_{k \geq 0} x^{2^k}$* , OEIS, <https://oeis.org/A168365>.
- [3] P. D. Hanna, *A168366: coefficient of x^n in the $(n+1)$ -st iteration of $\sum_{k \geq 0} x^{2^k}$* , OEIS, <https://oeis.org/A168366>.
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- [8] R. P. Stanley, *Enumerative Combinatorics, Volume 1*, second edition, Cambridge University Press, 2012.
- [9] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
- [10] V. Reshetnikov, *ProveIt: formal mathematics and reproducible research artifacts*, <https://github.com/VladimirReshetnikov/ProveIt>, repository inspected October 1, 2026.
- [11] V. Reshetnikov, *ProveIt repository*, research-report collection. *Binary Carry Structure in Iterated Lacunary Series*, October 1, 2026, especially Questions 13.2–13.3. <https://github.com/VladimirReshetnikov/ProveIt>. Exact path, commit and file blob are recorded in Section 26 and `SOURCE_AUDIT.txt`. The earlier report is an AI-assisted, unrefereed manuscript. [Write note, batch 114: this is Part I of the present report.]
- [12] S. Barbero, U. Cerruti and N. Murru, *On the operations of sequences in rings and binomial type sequences*, Ricerche di Matematica (2018), doi:10.1007/s11587-018-0389-5. <https://arxiv.org/abs/1805.11922>. The subsection on compositions of generating functions is the background reference for Hurwitz-series composition and inversion.
- [13] E. Rowland and R. Yassawi, *Automatic congruences for diagonals of rational functions*, Journal de théorie des nombres de Bordeaux **27** (2015), 245–288, doi:10.5802/jtnb.901. <https://jtnb.centre-mersenne.org/item/10.5802/jtnb.901.pdf>. Theorem 1.8 and Proposition 1.9 provide the standard kernel criterion and prime-power congruence discussed in Section 23; their rational-diagonal hypotheses are not assumed for the lacunary series here.